



KTU NOTES

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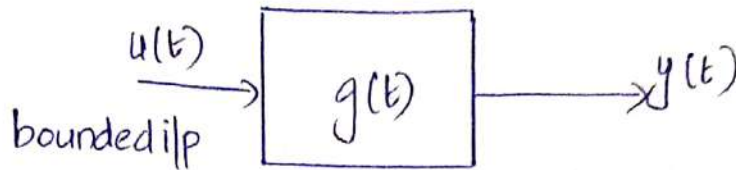
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MODULE 3

Bounded I/P Bounded O/P (BIBO)

If we give a bounded I/P, we should get a bounded O/P.



$$y(t) = u(t) \otimes g(t)$$

$$= \int_0^{\infty} u(t-\tau) g(\tau) d\tau$$

↳ convolution

$$|y(t)| = \left| \int_0^{\infty} u(t-\tau) g(\tau) d\tau \right|$$

$$|y(t)| \leq \int_0^{\infty} |u(t-\tau)| |g(\tau)| d\tau$$

$$\int_0^{\infty} |u(t)| dt \leq M$$

$$|y(t)| \leq \int_0^{\infty} M |g(\tau)| d\tau$$

$$|y(t)| \leq M \int_0^{\infty} |g(\tau)| d\tau$$

$$|y(t)| \leq N < \infty \text{ O/P is bounded.}$$

$$M \int_0^{\infty} |g(\tau)| d\tau \leq N$$

$$\int_0^{\infty} |g(\tau)| d\tau \leq \frac{N}{M}$$

$$\text{So } \int_0^{\infty} |g(\tau)| d\tau \leq Q$$

where $Q \leq \infty$.

★ If $g(t)$ is the impulse response:

$$\int_0^{\infty} g(t) dt \text{ — absolutely integrable}$$

$$\text{— finite value}$$

Location of poles

$$CLTF = \frac{G(s)}{1 + G(s)H(s)}$$

$$= \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

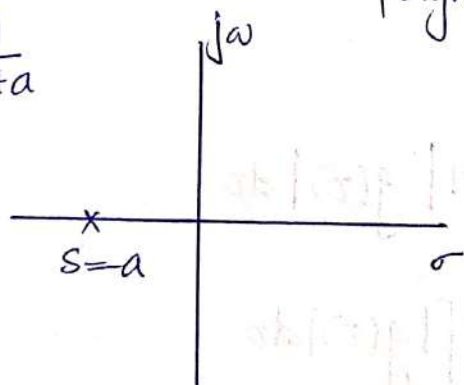
$$s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0 = 0 \quad \text{roots} \rightarrow \text{poles}$$

This eqn is known as characteristic eqn.

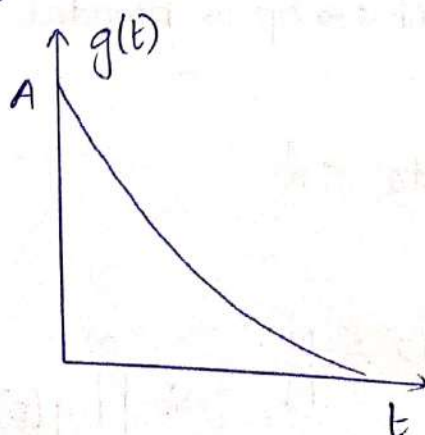
$$\text{or } 1 + G(s)H(s) = 0$$

Poles are roots of the characteristic equation (It is the denominator polynomial of CLTF).

• If $G(s) = \frac{A}{s+a}$

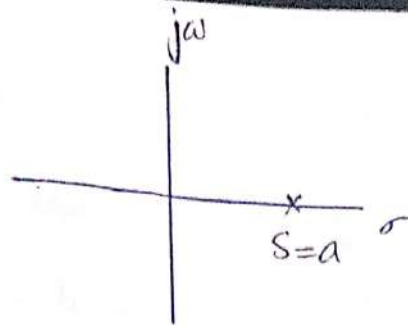


$$g(t) = A e^{-at}$$



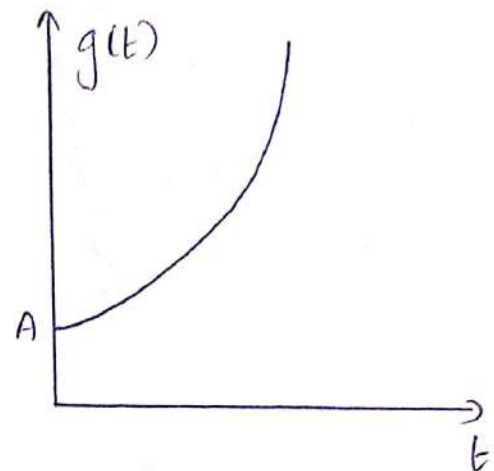
It is bounded
→ stable

$$G(s) = \frac{A}{s-a}$$

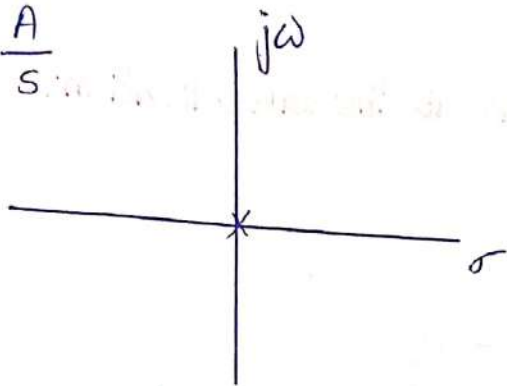


unbounded, unstable

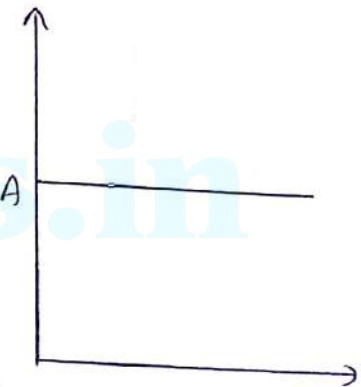
$$g(t) = Ae^{at}$$



$$G(s) = \frac{A}{s}$$



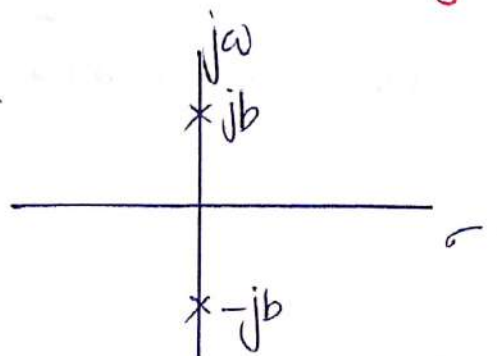
$$g(t) = A \text{ or } Au(t)$$



neither increasing nor decreasing
(maintaining constant amplitude)

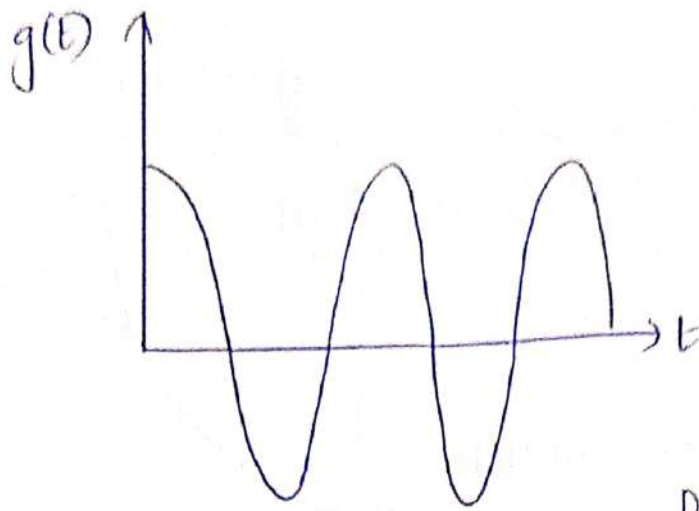
⇓
marginally stable

$$G(s) = \frac{A}{s+jb} + \frac{A}{s-jb}$$



$$g(t) = Ae^{-jbt} + Ae^{jbt}$$

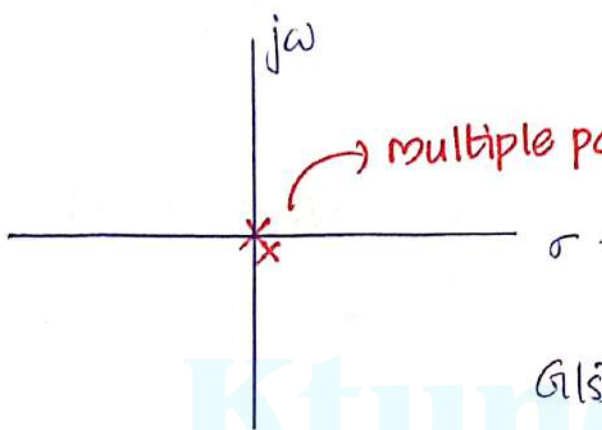
$$= 2A \cos bt$$



oscillatory response
(sustained oscillations)

Marginally stable

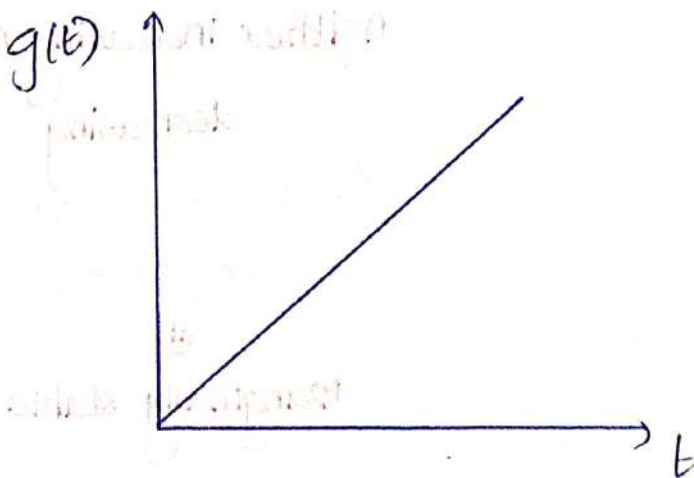
↓
no decay or exponential test increase



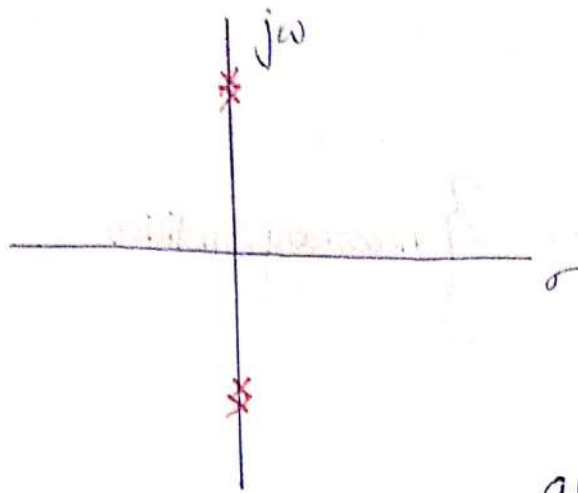
$$G(s) = \frac{A}{s^2}$$

both poles are @ origin.

$$g(t) = At$$



It is unstable ($\because$ it is unbounded)

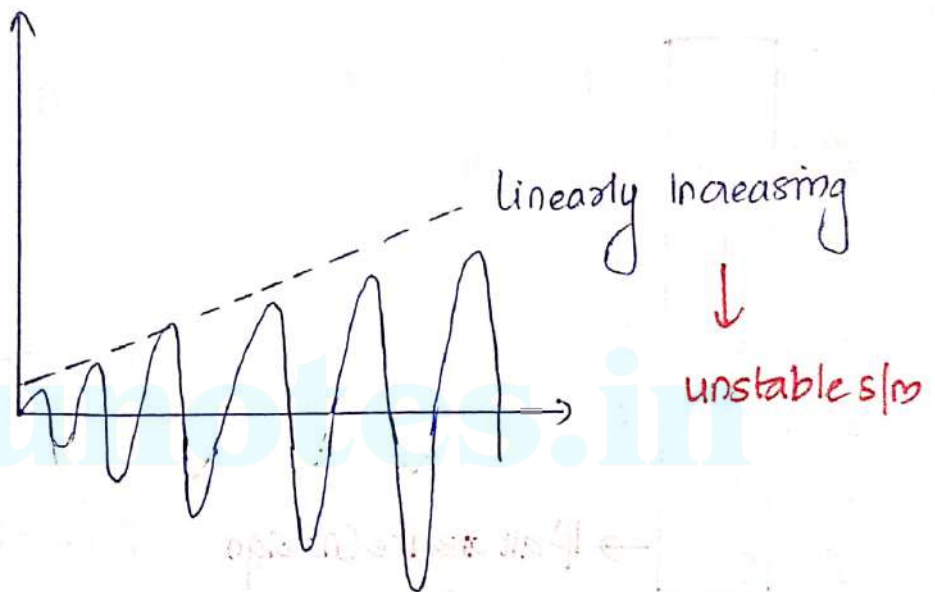


$$q(s) = \frac{A}{(s+jb)^2} + \frac{A}{(s-jb)^2}$$

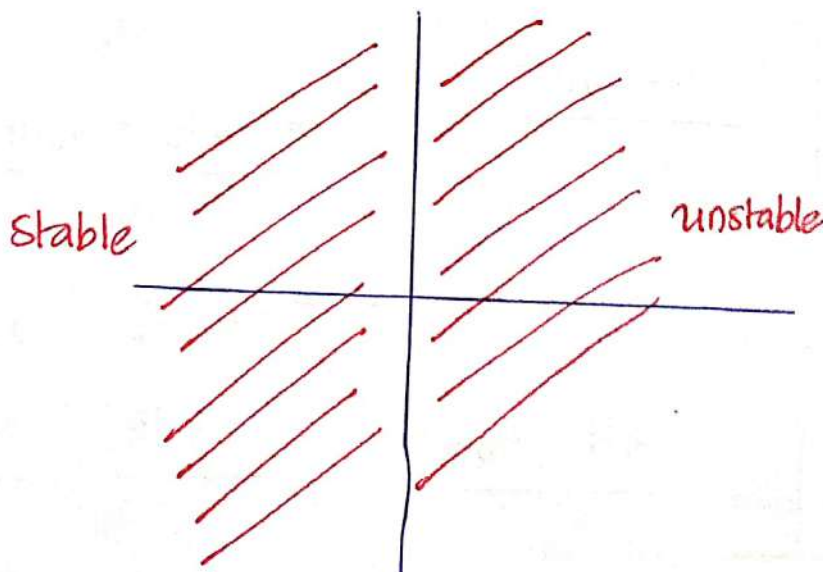
2 Poles are located at jb and $-jb$ resp.

$$q(t) = Ate^{-jbt} + Ate^{jbt}$$

$$= \underline{\underline{2A\cos bt}}$$



Conclusion:



↳ on the imaginary axis, it can be either marginally stable or unstable depending on the no. of poles.

$$\rightarrow s^n + a_{n-1}s^{n-1} + \dots + a_1s + a_0 = 0$$

For a s/m to be stable:

- (i) All coefficients of s should be ve } *necessary condition*
 (ii) No coefficients are zero

Routh Hurwitz Criterion

First step: formulating Routh Hurwitz array.

Eg: $a_6s^6 + a_5s^5 + a_4s^4 + a_3s^3 + a_2s^2 + a_1s + a_0 = 0$

$$s^6: \begin{array}{|c|c|c|c|} \hline a_6 & a_4 & a_2 & a_0 \\ \hline \end{array}$$

$$s^5: \begin{array}{|c|c|c|} \hline a_5 & a_3 & a_1 \\ \hline \end{array}$$

$$s^4: \begin{array}{|c|c|c|} \hline A & B & C \\ \hline \end{array}$$

$$s^3: \begin{array}{|c|c|} \hline D & E \\ \hline \end{array}$$

$$s^2: \begin{array}{|c|c|} \hline F & G \\ \hline \end{array}$$

$$s^1: \begin{array}{|c|} \hline H \\ \hline \end{array}$$

$$s^0: \begin{array}{|c|} \hline I \\ \hline \end{array}$$

$\rightarrow$ If all are ve (no sign changes) $\Rightarrow$ roots are on LHS of s and s/m is stable.

$$D = -1 \begin{vmatrix} a_5 & a_3 \\ A & B \end{vmatrix} = \frac{a_3A - a_5B}{A}$$

$$E = -1 \begin{vmatrix} a_5 & a_1 \\ A & C \end{vmatrix} = \frac{a_1A - a_5C}{A}$$

$$A = -1 \begin{vmatrix} a_6 & a_4 \\ a_5 & a_3 \end{vmatrix} = \frac{a_5a_4 - a_6a_3}{a_5}$$

$$= \frac{a_5a_4 - a_6a_3}{a_5}$$

$$B = -1 \begin{vmatrix} a_6 & a_2 \\ a_5 & a_1 \end{vmatrix} = \frac{a_5a_2 - a_6a_1}{a_5}$$

$$= \frac{a_5a_2 - a_6a_1}{a_5}$$

$$C = -1 \begin{vmatrix} a_6 & a_0 \\ a_5 & 0 \end{vmatrix} = \frac{a_5a_0}{a_5}$$

$$= \frac{a_5a_0}{a_5} = a_0$$

$$F = -1 \begin{vmatrix} A & B \\ D & E \end{vmatrix} = \frac{BD - AE}{D}$$

$$G = -1 \begin{vmatrix} A & C \\ D & 0 \end{vmatrix} = \frac{CD}{D} = \underline{\underline{C}}$$

$$H = -1 \begin{vmatrix} D & E \\ F & G \end{vmatrix} = \frac{EF - DG}{F}$$

$$I = -1 \begin{vmatrix} F & G \\ H & 0 \end{vmatrix} = \frac{GH}{H} = \underline{\underline{G}}$$

→ So if

First column

$\begin{cases} a_6 \\ a_5 \\ A \\ D \\ F \\ H \\ I \end{cases}$

all are +ve → stable.

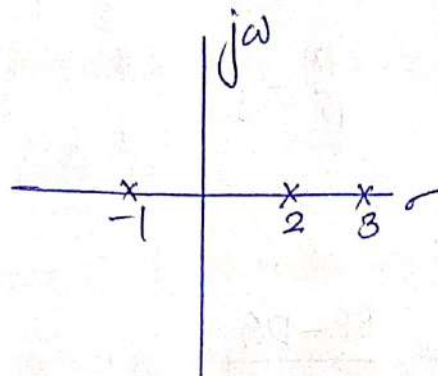
In case of sign changes:

no. of sign changes = no. of roots lying on the right hand side of the imaginary axis.

2 sign changes ⇒ 2 poles on the RHS.
⇒ unstable.

Q) A s/m with characteristic eqn $(s+1)(s-2)(s-3)=0$. Determine the stability of the s/m by determining the location of poles.

$$s = -1, 2, 3$$



unstable.

Using Routh Hurwitz criterion.

$$(s+1)(s-2)(s-3)=0$$

$$(s^2 - s - 2)(s-3)=0$$

$$s^3 - 3s^2 - s^2 + 3s - 2s + 6 = 0$$

$$s^3 - 4s^2 + s + 6 = 0$$

$$\begin{array}{cc|cc} s^3: & 1 & 1 & \\ s^2: & -4 & 6 & \\ s^1: & & & \end{array} \longrightarrow \begin{array}{cc|cc} s^3: & 1 & 1 & \\ s^2: & -2 & 3 & \\ s^1: & A & & \\ s^0: & B & & \end{array}$$

$$A = -1 \left| \begin{array}{cc} 1 & 1 \\ -2 & 3 \end{array} \right| = \frac{3+2}{2} = 2.5$$

$$B = -1 \left| \begin{array}{cc} A-2 & 3 \\ A & 0 \end{array} \right| = -1 \left| \begin{array}{cc} 2.5-2 & 3 \\ 2.5 & 0 \end{array} \right| = -1 \left| \begin{array}{cc} 0.5 & 3 \\ 2.5 & 0 \end{array} \right| = -1 \left(0 - 7.5 \right) = 7.5$$

$$= \frac{-1 \times 7.5}{2.5} = \underline{\underline{-3}}$$

The elements of the first column;

$$\begin{array}{c} 1 \\ -2 \\ 2.5 \\ 3 \end{array}$$

↖ sign change
↖ sign change

2 sign changes $\Rightarrow$ 2 poles on RHS.

The remaining pole is on LHS.

So s/m is unstable

Q) Using Routh criterion, determine the stability of the s/m represented by the characteristic eqn $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$. Comment on the location of the roots and determine whether the s/m is stable or not.

$$s^4 : 1 \quad 18 \quad 5$$

$$s^3 : 8 \quad 16$$

$$s^2 : A \quad B$$

$$s^1 : C$$

$$s^0 : D$$

$$A = -1 \left| \begin{array}{cc} 1 & 18 \\ 8 & 16 \end{array} \right|$$

$$\underline{\underline{8}}$$

$$= -1 (16 - 144)$$

$$\underline{\underline{8}}$$

$$= \frac{128}{8}$$

$$= \underline{\underline{16}}$$

$$C = -1 \left| \begin{array}{cc} 8 & 16 \\ A & B \end{array} \right| = -1 \left| \begin{array}{cc} 8 & 16 \\ 16 & 5 \end{array} \right|$$

$$\underline{\underline{16}}$$

$$= \frac{27}{162}$$

$$\underline{\underline{162}}$$

$$B = -1 \left| \begin{array}{cc} 1 & 5 \\ 8 & 0 \end{array} \right| = \frac{40}{8}$$

$$= \underline{\underline{5}}$$

$$D = -1 \left| \begin{array}{cc} 16 & 5 \\ 13.5 & 0 \end{array} \right| = \underline{\underline{5}}$$

13.5

No sign changes $\Rightarrow$ 0 poles on RHS
4 poles on LHS

Stable s/m

Q) Construct a Routh array and determine the stability of the s/m whose characteristic eqn is $s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s + 16 = 0$. Identify the location of the poles, determine whether it is a stable s/m or not.

$$s^6: 1 \quad 8 \quad 20 \quad 16$$

$$s^5: 2 \quad 12 \quad 16$$

$$s^4: A \quad B \quad C$$

$$s^3: D \quad E$$

$$s^2: F \quad G$$

$$s^1:$$

$$s^0:$$

$$A = -1 \left| \begin{array}{cc} 1 & 8 \\ 2 & 12 \end{array} \right| = \underline{\underline{2}}$$

$$B = -1 \left| \begin{array}{cc} 1 & 20 \\ 2 & 16 \end{array} \right| = \underline{\underline{12}}$$

$$C = -1 \left| \begin{array}{cc} 1 & 16 \\ 2 & 0 \end{array} \right| = \underline{\underline{16}}$$

$$D = -1 \begin{vmatrix} 2 & 12 \\ 2 & 12 \end{vmatrix} = \underline{\underline{0}}$$

$$E = -1 \begin{vmatrix} 2 & 16 \\ 2 & 16 \end{vmatrix} = \underline{\underline{0}}$$

$$F = -1 \begin{vmatrix} 2 & 12 \\ 5 & 0 \end{vmatrix} = \underline{\underline{12}}$$

$$G = -1 \begin{vmatrix} 2 & 16 \\ 5 & 0 \end{vmatrix} = \underline{\underline{16}}$$

Row of all zeroes should be an odd power of s so that the highest power of the eqn is even.

$$s^3: 0 \quad 0$$

Row of all zeros $\Rightarrow$ there can be roots/poles lying on imaginary axis

$$\text{Auxiliary eqn: } s^4 + 6s^2 + 8 = 0$$

- $\rightarrow$ Auxiliary eqn will always be an even power of s .
- $\rightarrow$ Auxiliary eqn will always be a factor of the characteristic eqn.
- $\rightarrow$ If we have auxiliary eqn is a slm, there are roots/poles lying on imag. axis (there is a chance).

$$s^4 + 6s^2 + 8 \quad \left) \begin{array}{r} s^2 + 2s + 2 \\ s^6 + 2s^5 + 8s^4 + 12s^3 + 20s^2 + 16s \\ s^6 + 0 + 6s^4 + 0 + 8s^2 + 16 \end{array}$$

$$\begin{array}{r} 2s^5 + 2s^4 + 12s^3 + 12s^2 + 16s + 16 \\ 2s^5 + 12s^3 + 0 + 16s + 16 \end{array}$$

$$2s^4 + 12s^2 + 16$$

$$2s^4 + 12s^2 + 16$$

$$\underline{\underline{0}}$$

$\rightarrow$ Roots of the auxiliary eqn will be the roots of characteristic eqn.

Differentiate the auxiliary eqn $A = s^4 + 6s^2 + 8 = 0$

$$\frac{dA}{ds} = 4s^3 + 12s = 0$$

The coefficients of this eqn replaces is taken to Routh array.

$$s^6: 1 \quad 8 \quad 20 \quad 16$$

$$s^5: 2 \quad 12 \quad 16$$

$$s^4: 2 \quad 12 \quad 16$$

$$s^3: 0 \quad 0$$

$$s^2: 4 \quad 12$$

$$s^1: I \quad J$$

$$s^0: K$$

$$s^0: L$$

$$I = -1 \left| \begin{array}{cc} 2 & 12 \\ 4 & 12 \end{array} \right| = \underline{\underline{6}}$$

$$J = -1 \left| \begin{array}{cc} 2 & 16 \\ 4 & 0 \end{array} \right| = \underline{\underline{16}}$$

$$K = -1 \left| \begin{array}{cc} 4 & 12 \\ 6 & 16 \end{array} \right| = \underline{\underline{\frac{8}{6}}}$$

$$L = -1 \left| \begin{array}{cc} 6 & 16 \\ \frac{8}{6} & 0 \end{array} \right| = \underline{\underline{\frac{16}{8/6}}}$$

No poles on Rhs but marginally stable

$$AE = s^4 + 6s^2 + 8 = 0$$

$$\text{let } s^2 = x$$

$$AE = x^2 + 6x + 8 = 0$$

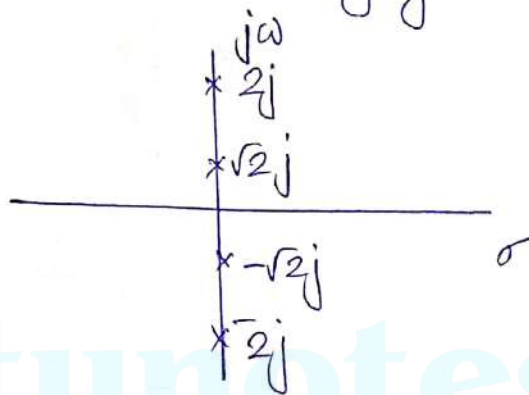
$$\Rightarrow x^2 + 4x + 2x + 8 = 0$$

$$\Rightarrow x(x+4) + 2(x+4) = 0$$

$$x = -2, -4$$

$$s = \pm \sqrt{2}j, \pm 2j$$

There are four roots/poles lying on the imaginary axis.



Out of 6 poles, 4 poles on imaginary axis and rest of the 2 poles is on LHS.

It is a marginally stable slm

Q) Construct the Routh array and determine the stability of the slm represented by the chara. eqn $s^5 + s^4 + 2s^3 + 2s^2 + 3s + 5 = 0$

$$s^5: 1 \quad 2 \quad 3$$

$$s^4: 1 \quad 2 \quad 5$$

$$s^3: A \quad B$$

$$s^2: C$$

$$A = -1 \begin{vmatrix} 1 & 2 \\ 1 & 2 \end{vmatrix} = \underline{\underline{0}}$$

$$B = -1 \begin{vmatrix} 1 & 3 \\ 1 & 5 \end{vmatrix}$$

$$= \underline{\underline{-2}}$$

$$C = -1 \left| \begin{array}{cc} 1 & 2 \\ 0 & -2 \end{array} \right|$$

$$0$$

Special case : first element of a row is 0.
 $S_3 : 0 \quad -2.$

$$S^5 : 1 \quad 2 \quad 3$$

$$S^4 : 1 \quad 2 \quad 5$$

$$S^3 : \epsilon \quad -2$$

$$S^2 : \frac{2\epsilon+2}{\epsilon} \quad D$$

$$S^1 : E$$

$$S^0 : F$$

$$C = \frac{2\epsilon+2}{\epsilon}$$

$$D = -1 \left| \begin{array}{cc} 1 & 5 \\ \epsilon & 0 \end{array} \right| = \underline{\underline{5}}$$

$$E = -1 \left| \begin{array}{cc} \epsilon & -2 \\ \frac{2\epsilon+2}{\epsilon} & D \end{array} \right|$$

$$\frac{2\epsilon+2}{\epsilon}$$

$$= -\epsilon D - \frac{4\epsilon-4}{\epsilon} \bigg/ \frac{2\epsilon+2}{\epsilon}$$

$$= - \frac{[\epsilon^2 D + 4\epsilon + 4]}{2\epsilon+2}$$

$$= - \frac{[5\epsilon^2 + 4\epsilon + 4]}{2\epsilon+2}$$

$$F = -1 \left| \begin{array}{cc} \frac{2\epsilon+2}{\epsilon} & 5 \\ \frac{-5\epsilon^2-4\epsilon-4}{2\epsilon+2} & 0 \end{array} \right| = \underline{\underline{5}}$$

$$\frac{-5\epsilon^2-4\epsilon-4}{2\epsilon+2}$$

Substituting $\epsilon = 0$;

$$\begin{array}{lcl} s^5: & 1 & 2 \quad 3 \\ s^4: & 1 & 2 \quad 5 \\ s^3: & 0 & -2 \\ s^2: & +\infty & 5 \\ s^1: & -2 & \\ s^0: & 5 & \end{array}$$

Two sign changes $\Rightarrow$ 2 poles on RHS
Unstable s/m

Qn: Determine the stability of the system $s^6 + s^5 + 3s^4 + 3s^3 + 3s^2 + 2s + 1$.

$$\begin{array}{lcl} s^6: & 1 & 3 \quad 3 \quad 1 \\ s^5: & 1 & 3 \quad 2 \\ s^4: & A & B \quad C \\ & 0 & 1 \quad 1 \\ s^3: & D & E \\ s^2: & F & G \\ s^1: & H & \\ s^0: & I & \end{array}$$

$$A = -1 \left| \begin{array}{cc} 1 & 3 \\ 1 & 3 \end{array} \right|$$

$$= 0$$

$$B = -1 \left| \begin{array}{cc} 1 & 3 \\ 1 & 2 \end{array} \right|$$

$$= 1$$

$$C = -1 \left| \begin{array}{cc} 1 & 1 \\ 1 & 0 \end{array} \right|$$

$$= 1$$

$$D = -1 \begin{vmatrix} 1 & 3 \\ \varepsilon & 1 \end{vmatrix}$$

$$= \frac{3\varepsilon - 1}{\varepsilon}$$

$$E = -1 \begin{vmatrix} 1 & 2 \\ \varepsilon & 1 \end{vmatrix} = \frac{2\varepsilon - 1}{\varepsilon}$$

$$F = -1 \begin{vmatrix} \varepsilon & 1 \\ \frac{3\varepsilon - 1}{\varepsilon} & \frac{2\varepsilon - 1}{\varepsilon} \end{vmatrix} = \frac{\frac{3\varepsilon - 1}{\varepsilon} - (2\varepsilon - 1)}{\frac{3\varepsilon - 1}{\varepsilon}}$$

$$= \frac{3\varepsilon - 1 - 2\varepsilon^2 + \varepsilon}{3\varepsilon - 1}$$

$$= \frac{-2\varepsilon^2 + 4\varepsilon - 1}{3\varepsilon - 1}$$

$$G = -1 \begin{vmatrix} \varepsilon & 1 \\ \frac{3\varepsilon - 1}{\varepsilon} & 0 \end{vmatrix} = \frac{\frac{3\varepsilon - 1}{\varepsilon}}{\frac{3\varepsilon - 1}{\varepsilon}} = 1$$

$$H = -1 \begin{vmatrix} \frac{3\varepsilon - 1}{\varepsilon} & \frac{2\varepsilon - 1}{\varepsilon} \\ -2\varepsilon^2 + 4\varepsilon - 1 & 1 \end{vmatrix} = \frac{-\left[\frac{3\varepsilon - 1}{\varepsilon} \cdot \frac{(-2\varepsilon^2 + 4\varepsilon - 1) \times 2\varepsilon - 1}{\varepsilon} \right]}{-2\varepsilon^2 + 4\varepsilon - 1}$$

$$\begin{aligned}
 &= - \left[\frac{3s-1}{s} - \left(\frac{-4s^3 + 8s^2 - 2s + 2s^2 - 4s + 1}{3s^2 - s} \right) \right] \\
 &\quad \frac{-2s^2 + 4s - 1}{3s - 1} \\
 &= - \left[\frac{1}{s} \left(\frac{(3s-1)^2 + 4s^3 - 8s^2 + 2s - 2s^2 + 4s - 1}{3s - 1} \right) \right] \\
 &\quad \frac{-2s^2 + 4s - 1}{3s - 1} \\
 &= - \frac{1}{s} \left[\frac{9s^2 + 1 - 6s + 4s^3 - 8s^2 + 2s - 4s - 2s^2 - 1}{-2s^2 + 4s - 1} \right] \\
 &= \frac{4s^2 - s}{2s^2 - 4s + 1}
 \end{aligned}$$

$$P = +1$$

$$s^6: 1 \quad 3 \quad 3 \quad 1$$

$$s^5: 1 \quad 3 \quad 2$$

$$s^4: 0 \quad 1 \quad 1$$

$$s^3: -\infty \quad -\infty$$

$$s^2: 1 \quad 1$$

$$s^1: 0 \rightarrow \text{row of all zeros}$$

$$s^0: 1$$

Two sign changes $\Rightarrow$ Two poles on the RHS.

Unstable s/m

Auxiliary eqn is $s^2 + 1 = 0$

$$s = \pm j$$

Two roots are located on the imaginary axis.

Remaining 2 roots on the LHS.

HW

Qn: Determine the stability of the s/m $s^5 + 4s^4 + 8s^3 + 8s^2 + 7s + 4 = 0$.

Determine the range of k for the stability of unity feedback s/m whose open loop transfer fn is $G(s) = \frac{k}{s(s+1)(s+2)}$

$$H(s) = 1$$

$$G(s) = \frac{k}{s(s+1)(s+2)}$$

$$s(s+1)(s+2) = 0$$

$$(s^2+s)(s+2) = 0$$

$$s^3 + 2s^2 + s^2 + 2s = 0$$

$$s^3 + 3s^2 + 2s = 0$$

$$CLTF = \frac{\frac{k}{s(s+1)(s+2)}}{1 + \frac{k}{s(s+1)(s+2)}} = \frac{k}{k + s(s+1)(s+2)}$$

$$k + s(s+1)(s+2) = 0$$

$s^3 + 3s^2 + 2s + k = 0$ is the characteristic eqn.

$$s^3: 1 \quad 2$$

$$s^2: 3 \quad k$$

$$s^1: A$$

$$s^0: B$$

$$A = -1 \left| \begin{array}{cc} 1 & 2 \\ 3 & k \end{array} \right| = \frac{6-k}{3}$$

$$B = -1 \left| \begin{array}{cc} 3 & k \\ \frac{6-k}{3} & 0 \end{array} \right| = \frac{k}{\frac{6-k}{3}}$$

$$s^3: 1 \quad 2$$

$$s^2: 3 \quad k$$

$$s^1: \frac{6-k}{3}$$

$$s^0: k$$

Range is $0 < k < 6$
open loop gain

Qn: The OLTF of a unity feedback s/m is $G(s) = \frac{k}{(s+2)(s+4)(s^2+6s+25)}$

Using Routh Hurwitz criterion, discuss the stability of the cl closed loop s/m as a function of k . Also, determine value of k which will cause sustained oscillations in the CL s/m; what are the corresponding oscillating frequencies?

$$H(s) = 1$$

$$CLTF = \frac{k}{(s+2)(s+4)(s^2+6s+25)}$$

$$1 + \frac{k}{(s+2)(s+4)(s^2+6s+25)}$$

$$\text{Characteristic eq} = (s+2)(s+4)(s^2+6s+25) + k = 0$$

$$(s^2+6s+8)(s^2+6s+25) + k = 0$$

$$s^4 + 6s^3 + 25s^2 + 6s^3 + 36s^2 + 150s + 8s^2 + 48s + 200 + k = 0$$

$$s^4 + 12s^3 + 69s^2 + 198s + 200 + k = 0$$

$$s^4: 1 \quad 69 \quad 200+k$$

$$s^3: 12 \quad 198$$

$$s^2: A \quad B$$

$$s^1: 0$$

$$s^0: 200+k$$

$$A = -1 \left| \begin{array}{cc} 1 & 69 \\ 12 & 198 \end{array} \right| = \frac{52.5}{12}$$

$$B = -1 \left| \begin{array}{cc} 1 & 200+k \\ 12 & 0 \end{array} \right| = \frac{2400+12k}{12} = \underline{\underline{200+k}}$$

$$C = -1 \left| \begin{array}{cc} 12 & 198 \\ 52.5 & 200+k \end{array} \right|$$

$$= \frac{10395 - 2400 - 12k}{52.5} = \frac{7995 - 12k}{52.5}$$

$$= \underline{\underline{152.2857 - 0.22857}}$$

For the s/m to be stable;

$$\frac{7995 - 12k}{52.5} > 0 \quad \text{and} \quad 200 + k > 0$$

$$k < 666.25$$

$$k > -200$$

$$\underline{\underline{-200 < k < 666.25}}$$

For sustained oscillations, there should be a row of all zeroes.

$$\frac{7995 - 12k}{52.5} = 0$$

$$\underline{\underline{k = 666.25}}$$

$$AE : 52.5s^2 + 200 + k = 0$$

$$52.5s^2 + 866.25 = 0$$

$$s^2 = -16.5$$

$$\underline{\underline{s = \pm 16.5j \pm 4.06j}}$$

$$\omega = 4.06 \text{ rad/s}$$

Tutorial Qn: The OLTF of a unity FB s/m is $G(s) = \frac{k(s+1)}{s^3+as^2+2s+1}$. Determine the value of k and a so that the s/m oscillates at a frequency of 2 rad/s.

$$\begin{aligned} \text{CLTF} &= \frac{\frac{k(s+1)}{s^3+as^2+2s+1}}{1 + \frac{k(s+1)}{s^3+as^2+2s+1}} = \frac{k(s+1)}{s^3+as^2+2s+1 + ks + k} \\ &= \frac{ks + s}{s^3+as^2 + (2+k)s + (1+k)} \end{aligned}$$

$$\text{chara. eqn : } s^3 + as^2 + (2+k)s + (1+k) = 0$$

$$s^3: 1 \quad 2+k$$

$$s^2: a \quad 1+k$$

$$s^1: A$$

$$s^0: B$$

$$A = -1 \left| \begin{array}{cc} 1 & 2+k \\ a & 1+k \end{array} \right|$$

$$= \frac{2a + ak - 1 - k}{a}$$

$$= \frac{(2a-1)k + (2a-1)}{a}$$

$$B = -1 \left| \begin{array}{cc} a & 1+k \\ \frac{(2a-1)k + (2a-1)}{a} & 0 \end{array} \right|$$

$$= \frac{(2a-1)k + (2a-1)}{a}$$

$$= \underline{\underline{1+k}}$$

For sustained oscillations ;

$$\frac{(a-1)k + (2a-1)}{a} = 0$$

$$(a-1)k = 1 - 2a$$

$$k = \frac{1-2a}{a-1}$$

Root locus

$$G(s) \cdot H(s) = \frac{k(s+z_1)}{s(s+p_1)(s+p_2)} \quad (\text{Open loop transfer function})$$

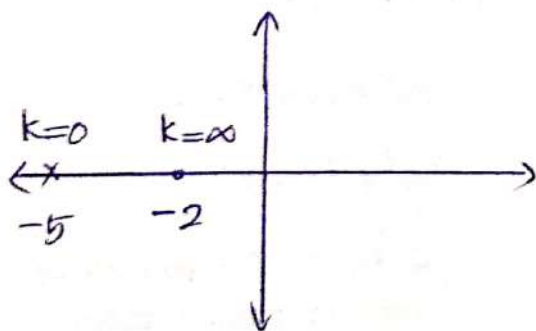
$$CLTF = \frac{k(s+z_1)}{s(s+p_1)(s+p_2) + k(s+z_1)}$$

$$\text{Characteristic eqn} \Rightarrow s(s+p_1)(s+p_2) + k(s+z_1) = 0$$

$$k=0 \Rightarrow s(s+p_1)(s+p_2) = 0 \quad s=0, -p_1, -p_2 \quad (\text{poles of OLTF})$$

$$k=\infty \Rightarrow s(s+p_1)(s+p_2) + k(s+z_1) = 0 \quad s = -z_1 \quad (\text{zeros of OLTF})$$

$$G(s) \cdot H(s) = \frac{k(s+2)}{s+5}$$



★ Poles of OLTF $\equiv$ Poles of CLTF $\equiv$ Roots of CE $[k=0]$

Zeros of OLTF $\equiv$ Poles of CLTF $\equiv$ Roots of CE $[k=\infty]$

$$* CLTF = \frac{G(s)}{1 + G(s)H(s)}$$

$$1 + G(s)H(s) = 0$$

$$G(s) \cdot H(s) = -1$$

$$|G(s)H(s)| = 1$$

magnitude condition

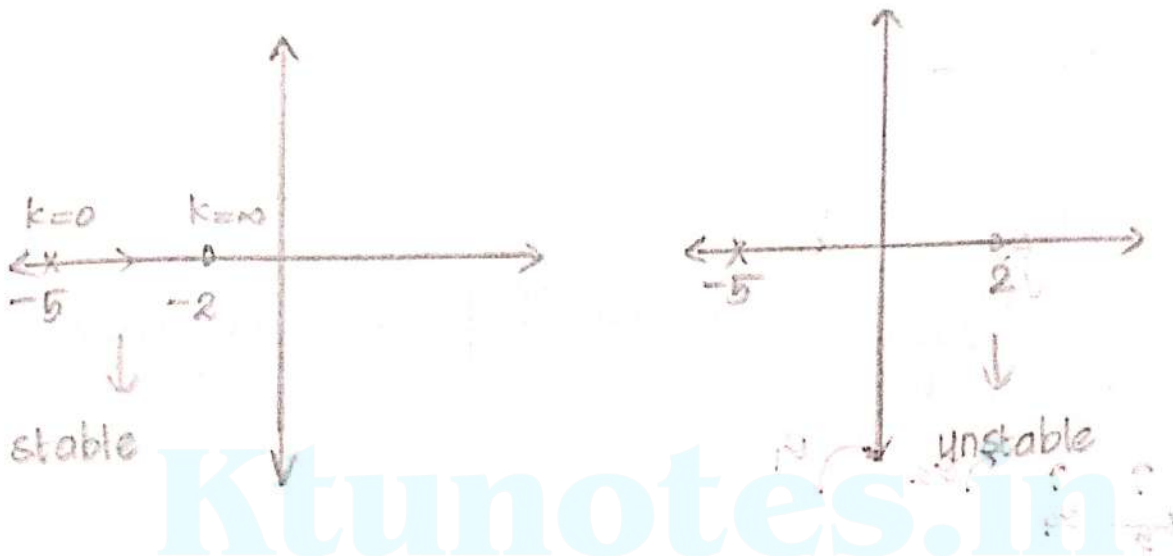
$$\angle G(s)H(s) = \pm 180^\circ (2q+1)$$

$$q = 0, 1, 2, 3$$

→ Angle condition

Any point on the root locus should satisfy the magnitude and angle condition.

Eg:



$$*G(s)H(s) = \frac{k(s+z_1)(s+z_2)(s+z_3)\dots}{(s+p_1)(s+p_2)(s+p_3)\dots}$$

General form

Magnitude

$$\frac{k |s+z_1| |s+z_2| |s+z_3| \dots}{|s+p_1| |s+p_2| |s+p_3| \dots} = 1$$

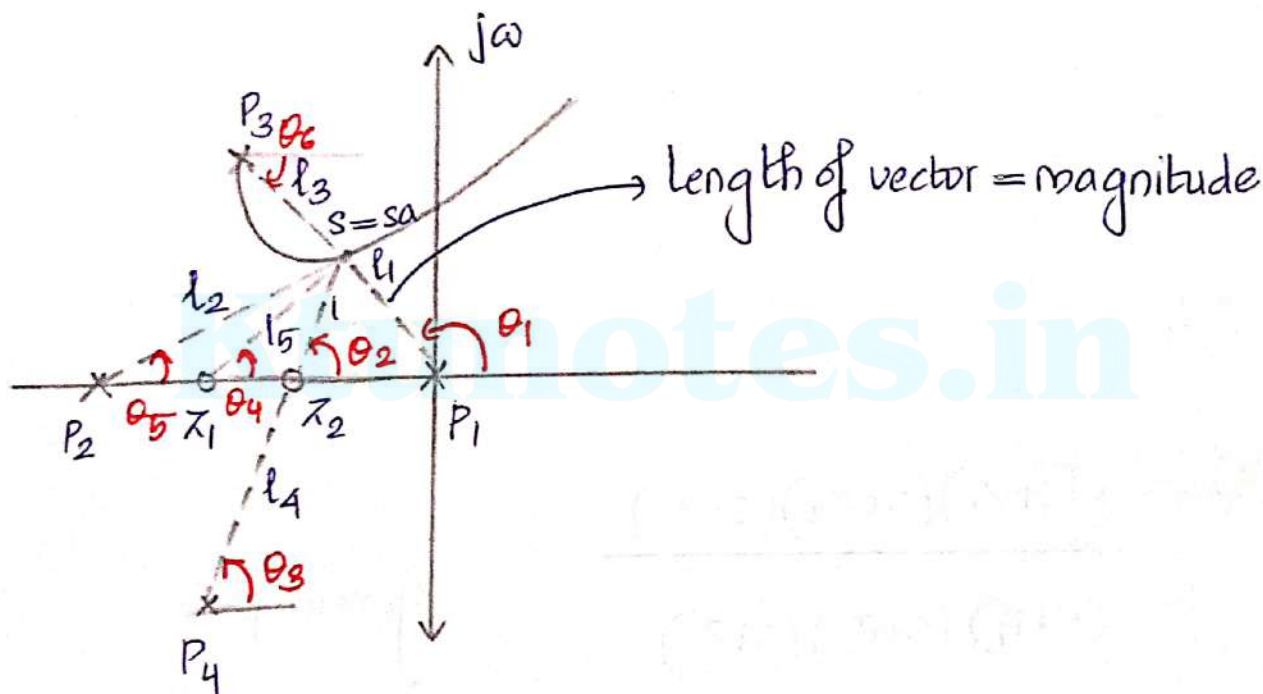
$$k = \frac{|s+p_1| |s+p_2| |s+p_3| \dots}{|s+z_1| |s+z_2| |s+z_3| \dots}$$

Angle

$$\angle(s+z_1) + \angle(s+z_2) + \angle(s+z_3) + \dots - [\angle(s+p_1) + \angle(s+p_2) + \angle(s+p_3)]$$

ie { sum of angles in the numerator - sum of angles in the denominator }

$$= \underline{\underline{\pm 180^\circ (2q+1)}}$$



$$\underline{s = s_a}$$

$$K = \frac{|s_a + p_1| |s_a + p_2| |s_a + p_3| \dots}{|s_a + z_1| |s_a + z_2| |s_a + z_3| \dots} \quad (\text{magnitude})$$

$$\underline{\underline{K = \frac{l_1 l_2 l_3 l_4}{l_5 l_6}}}$$

Magnitude condition

k at any point $s = s_a$ on root locus =

Product of the length of the vectors drawn from O.L. poles to the point on the Root locus

Product of the length of the vectors drawn from open loop zeros to the point on the root locus.

At $s = s_a$

$$\angle(s_a + z_1) + \angle(s_a + z_2) + \angle(s_a + z_3) \dots - \left[\angle(s_a + p_1) + \angle(s_a + p_2) + \angle(s_a + p_3) \dots \right] = \pm 180^\circ (2q+1)$$

Angle condition

Sum of angles of vectors drawn from open loop zeros to the point on root locus - sum of angles of vectors drawn from open loop poles to point on root locus = $\pm 180^\circ (2q+1)$

$$\theta_3 + \theta_4 - [\theta_1 + \theta_2 + \theta_5 + \theta_6] = \pm 180^\circ (2q+1)$$

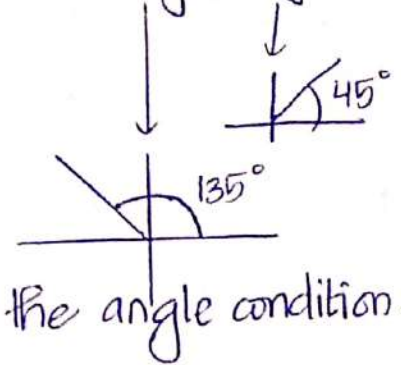
Q: Find the system gain so that the point $s = -3 + j3$ lies on the root locus, $G(s) = \frac{k}{s(s+6)}$

$$G(s)H(s) = \frac{k}{s(s+6)}$$

$$\angle G(s)H(s) \Big|_{s=-3+j3} = \angle \frac{k}{(-3+j3)(-3+j3+6)} = \angle \frac{k}{(-3+j3)(3+j3)}$$

$$= 0^\circ - [135^\circ + 45^\circ]$$

$$= -180^\circ$$



It is satisfying the angle condition.

$$|G(s)H(s)|_{s=-3+j3} = 1 \rightarrow \text{Magnitude condition.}$$

$$|G(s)H(s)| = \frac{k}{\sqrt{18} \cdot \sqrt{18}} = \frac{k}{18}$$

$$\frac{k}{18} = 1$$

$$\Rightarrow \underline{k=18}$$

Rules for constructing root locus diagram

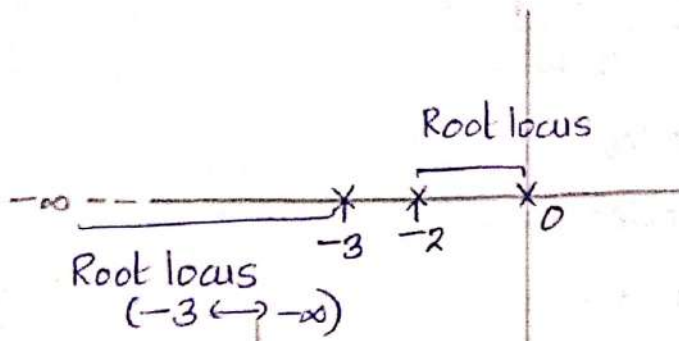
- Root locus will always be symmetrical to the real axis.
- The no. of roots of locus branches completely depends on the no. of poles and zeros.

n - poles
 m - zeros

if

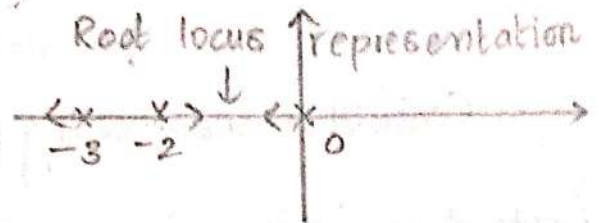
- Identification of root locus branches on real axis.

Identification of root locus branches on the real axis.

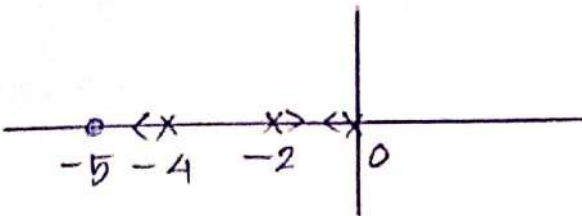
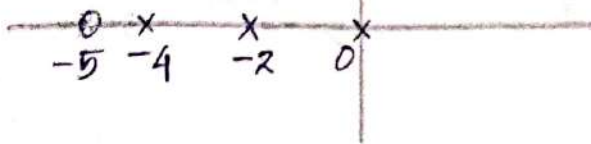


Right side of -2 has odd (1) number of poles. So it is root locus point whereas -3 to -2 is not a root locus branch.

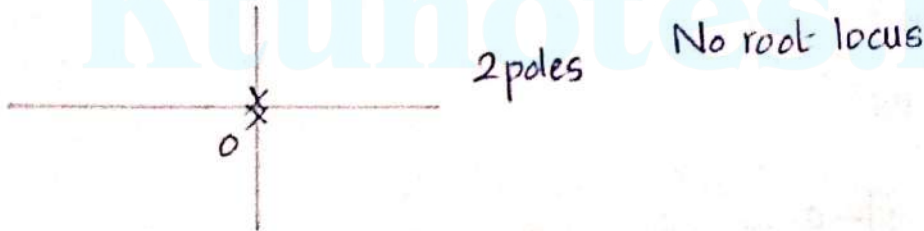
Root locus representation



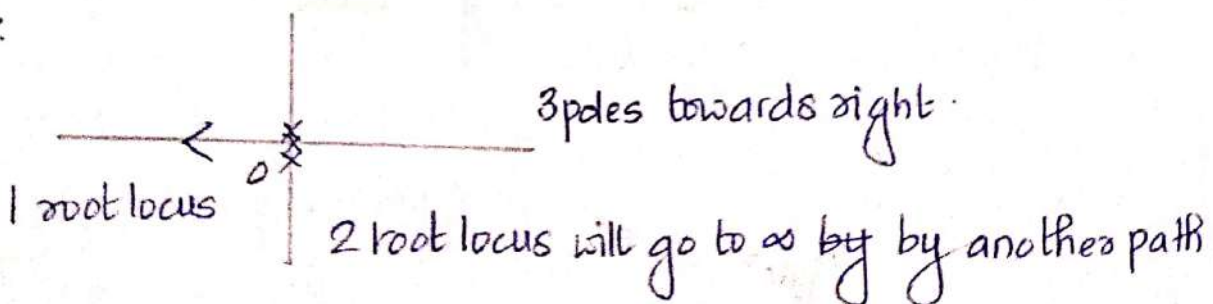
Q:



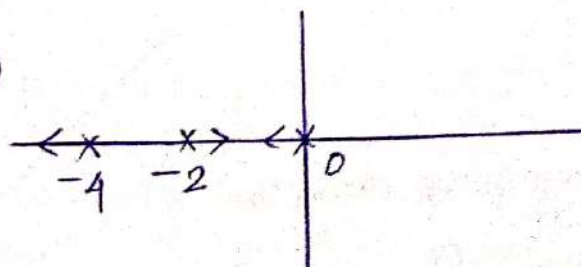
Q:



Q:



Q: $G(s) \cdot H(s) = \frac{K}{s(s+2)(s+4)}$



Asymptotes

$$\text{angle of asymptotes} = \frac{\pm 180^\circ (2q+1)}{n-m}$$

n - no. of poles

m - no. of poles zeros

$q = 0, 1, 2, 3, \dots$

Centroid is the meeting point of asymptotes on real axis.

$$\text{Centroid} = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$$

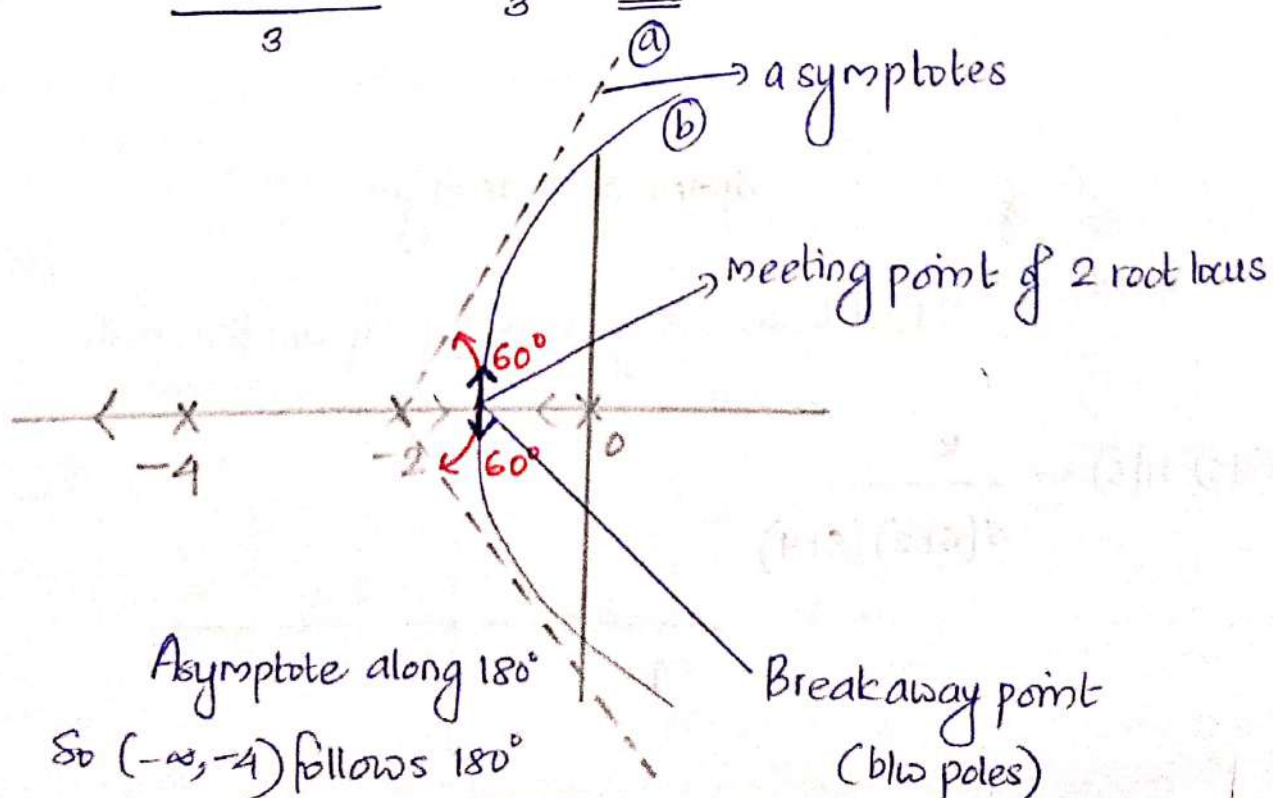
Eg: $n=3, m=0$;

When $q=0$, $\theta = \frac{\pm 180^\circ}{3} = \pm 60^\circ$

$q=1$, $\theta = \pm 180^\circ$

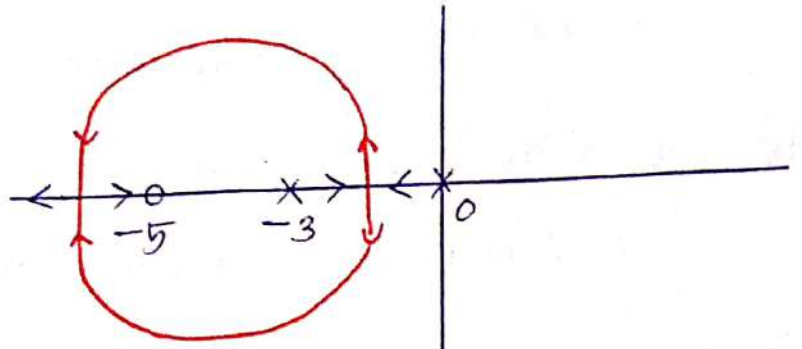
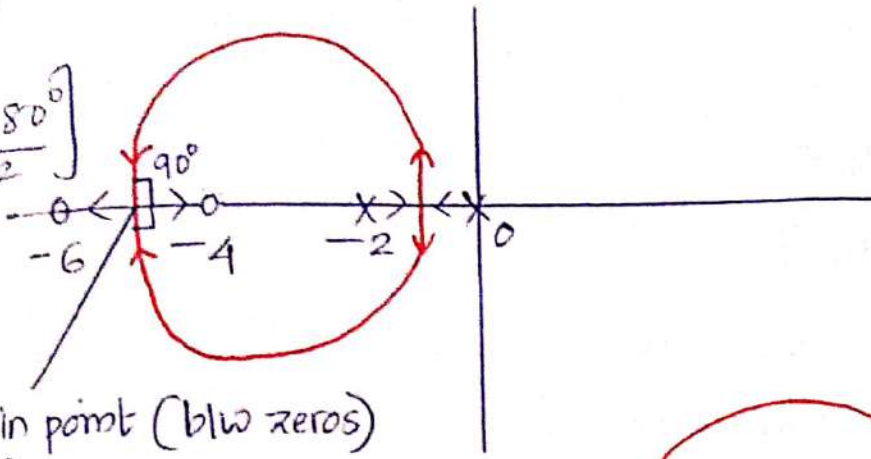
$q=2$, $\theta = \pm 300^\circ$

$$\text{Centroid} = \frac{(0-2-4)-0}{3} = -\frac{6}{3} = -2$$



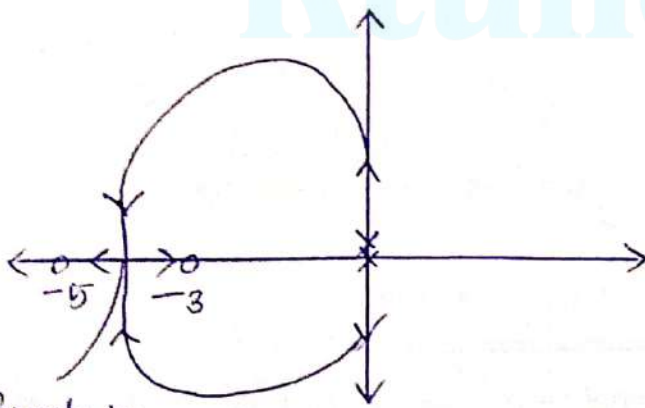
Eg:

$$\left[\frac{\pm 180^\circ}{2} \right]$$

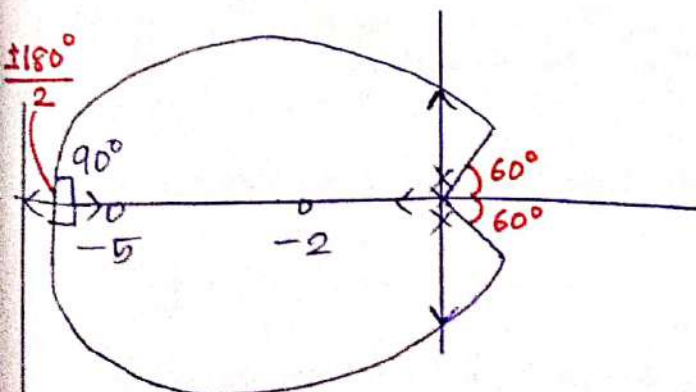


→ Angle at which root locus is getting separated from the meeting point = $\frac{\pm 180^\circ}{n}$.

n — no. of root locus branches meeting.



$$\frac{+180^\circ}{2} = \underline{\underline{\pm 90^\circ}}$$



$$\frac{+180^\circ}{3} = \underline{\underline{\pm 60^\circ}}$$

→ 2 root locus branches joining

1) Characteristic equation

$$B(s) + kA(s) = 0$$

$$k = \frac{-B(s)}{A(s)}$$

$$\frac{dk}{ds} = 0 \text{ — Find the roots}$$

(multiple roots)

Subs. root in characteristic eq, Find value of k .

(If k is +ve and real, break away pt)

$$Q_n) G(s)H(s) = \frac{k}{s(s+2)(s+4)}$$

$$C.E \Rightarrow s(s+2)(s+4) + k = 0$$

$$\Rightarrow s^3 + 6s^2 + 8s + k = 0$$

$$k = -(s^3 + 6s^2 + 8s)$$

$$\frac{dk}{ds} = 0$$

$$\frac{dk}{ds} = 3s^2 + 12s + 8 = 0$$

$$\text{roots} = -0.845, -3.154$$

Break away point must be b/w 0 and -2, so, it will be -0.845.

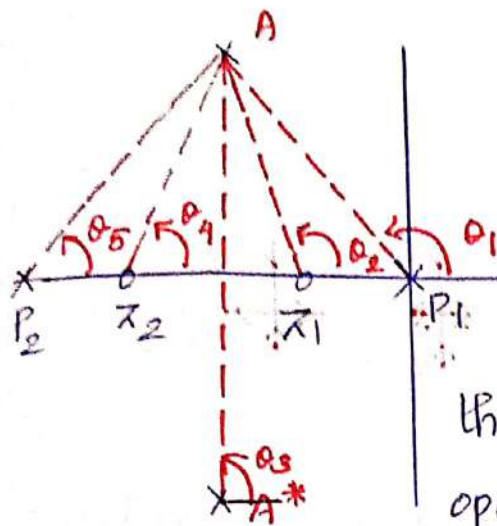
$$s = -0.845$$

$$k = -[(-0.845)^3 + 6(-0.845)^2 + 8(-0.845)]$$
$$= \underline{\underline{3.07}}$$

$$s = -3.154$$

$$k = \underline{\underline{-3.07}}$$

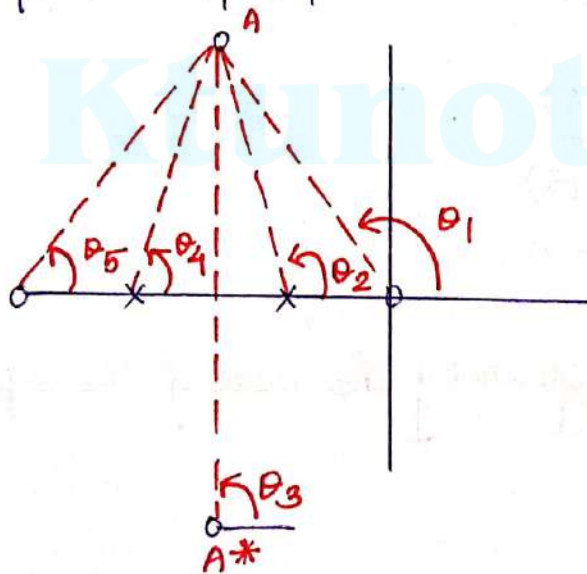
2) To find the angle of departure and angle of arrival (Complex poles)



Angle of departure from complex pole $A = 180^\circ - [\text{sum of angles of vectors drawn from all open loop poles to the complex poles}] + [\text{sum of the angles of the vectors drawn from the all open loop zeros to the complex pole}]$

$$= 180^\circ - (\theta_1 + \theta_3 + \theta_5) + (\theta_4 + \theta_2)$$

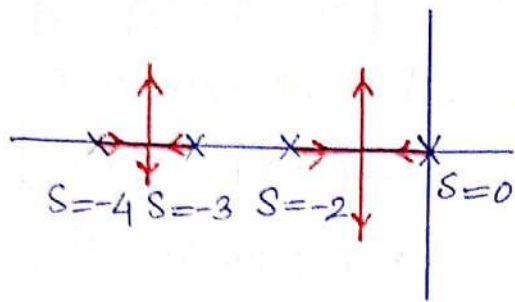
Angle of departure of complex conjugate pole $A^* = -\text{angle of departure from complex pole } A$.



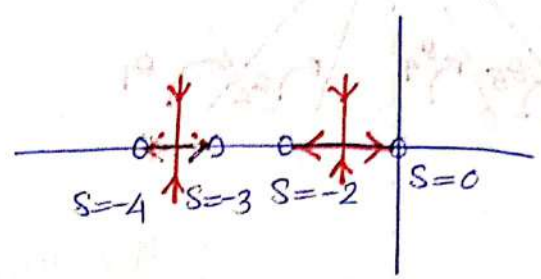
Angle of arrival at complex zero $A = 180^\circ - [\text{sum of angles of vectors drawn from all other open loop zeros to the complex zeros}] + [\text{sum of the angles of vectors drawn from all open loop poles to the complex zeros}]$

$$= 180^\circ - [\theta_1 + \theta_3 + \theta_5] + [\theta_4 + \theta_2]$$

Break away | Break in points



$k=0$ If there is a root locus branch between two poles on real axis, we'll have break away point.



If there is a root locus branch between two zeros on real axis, we will have a break in point

Characteristic eqn $B(s) + kA(s) = 0$

$$k = \frac{-B(s)}{A(s)}$$

k - positive and real

💡 Break away | break in point is given by the roots of the eqn $\frac{dk}{ds} = 0$. Substitute value of roots in characteristic equation and find the value of k .

Qn: Find break away and break in points for unity feedback s/s

$$G(s) = \frac{k}{s(s+2)(s+4)}$$

$$\text{characteristic equation} = s(s+2)(s+4) + k = 0$$

$$s^3 + 6s^2 + 8s + k = 0$$

$$k = -s^3 - 6s^2 - 8s$$

$$\frac{dk}{ds} = -3s^2 - 12s - 8$$

$$\frac{dk}{ds} = 0 \Rightarrow 3s^2 + 12s + 8 = 0$$

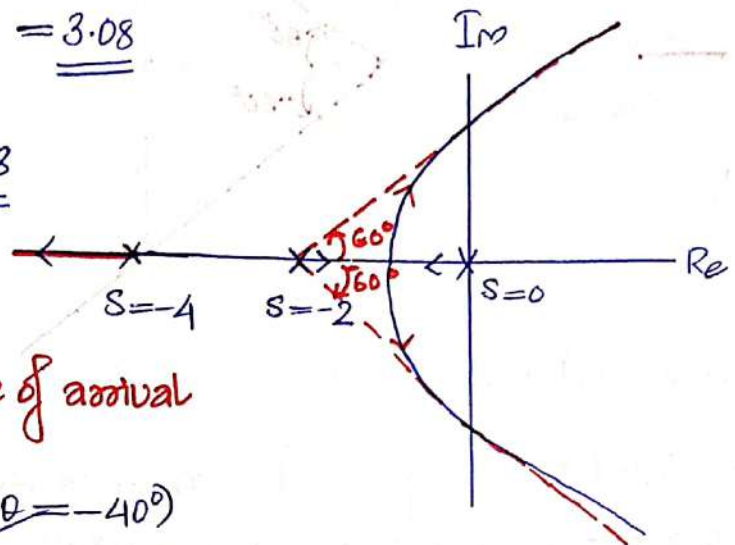
$$\text{roots} \rightarrow s = -0.845, -3.15$$

Substitute s in characteristic eqn and find k

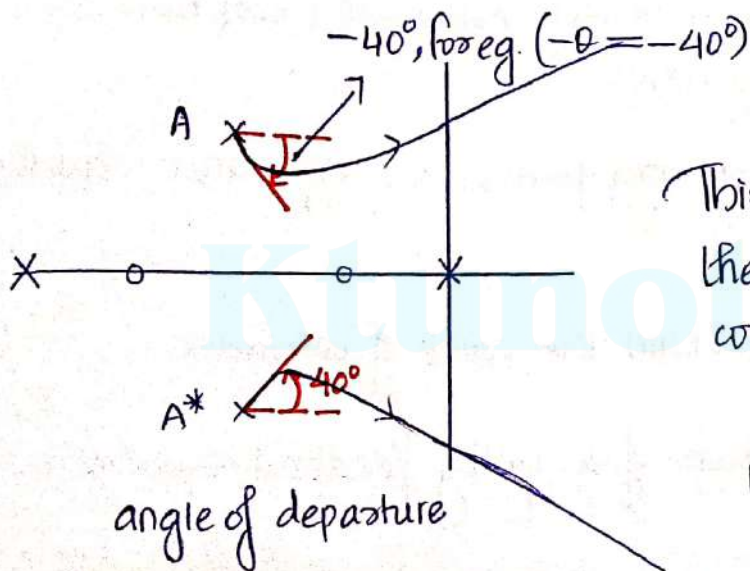
$$\text{When } s = -0.845 \quad k = -(-0.845)^3 - 6(-0.845)^2 - 8(-0.845)$$

$$\text{When } s = -3.15, \quad k = 3.08$$

$$\underline{k = -3.08}$$



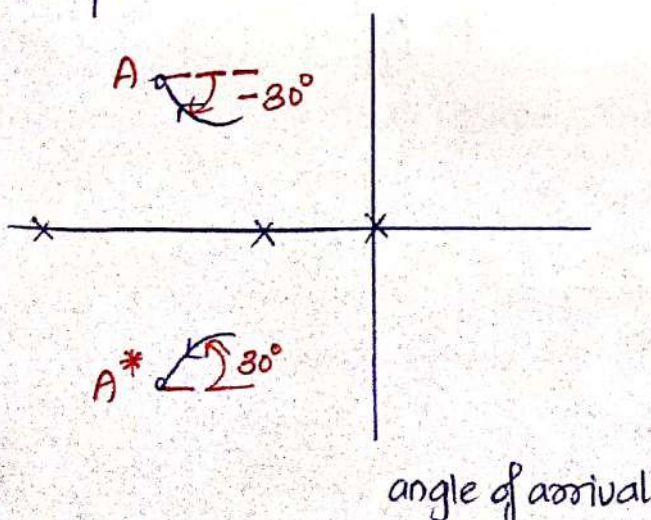
Angle of departure and angle of arrival



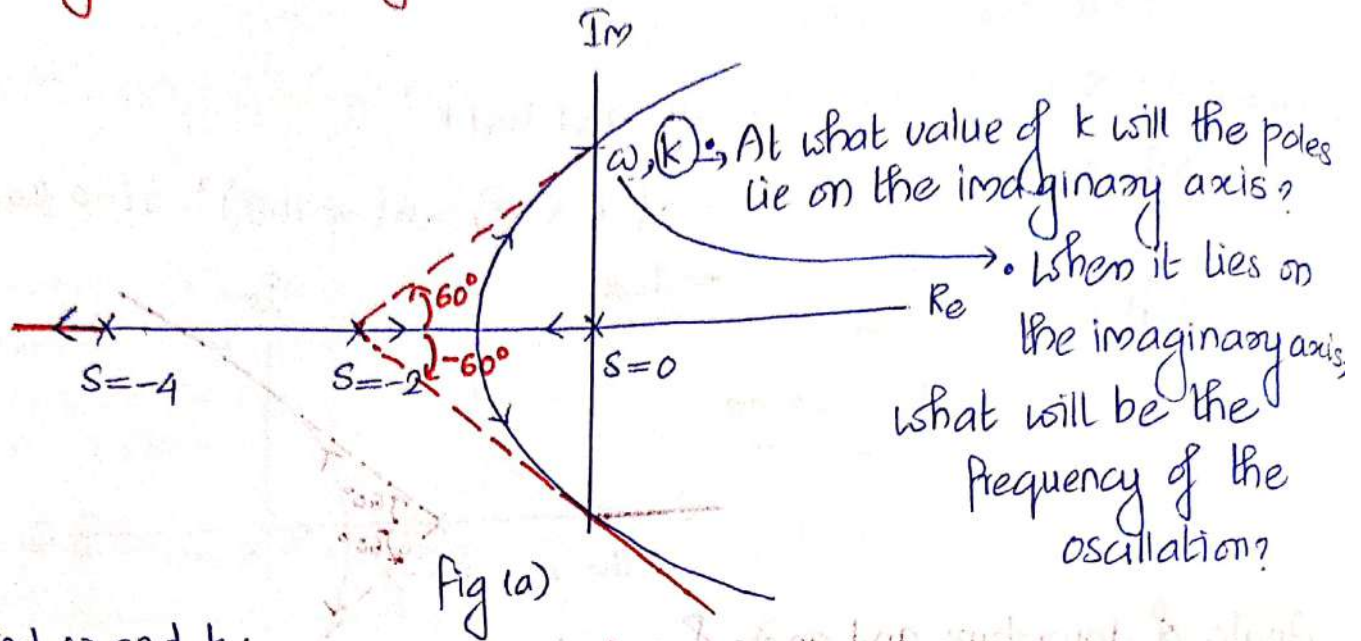
This rule is applicable only when there is a complex pole and its complex conjugate pair.
or

When there are complex zeros in the system.

→ Angle of departure needs to be computed only when there are complex poles in the s/m and angle of arrival when complex zeros are present.



Point of intersection of root locus with imaginary axis.



To find ω and k :

→ Substitute $s = j\omega$ in the characteristic equation (because there are no real parts on the imaginary axis).

Then, separate the real and imaginary parts in the eqn and equate to zero.

→ Solve these two eqns to obtain the value of ω and k .

Qn) The open loop transfer function of a unity feedback system is

$$G(s) = \frac{k}{s(s+2)(s+4)}$$

Find the point where the root locus crosses the imaginary axis. Also find the value of k at that point.

Refer fig (a).

$$\text{Ch eqn} \Rightarrow s(s+2)(s+4) + k = 0$$

$$s^3 + 6s^2 + 8s + k = 0$$

$$\text{Subs } s = j\omega;$$

$$(j\omega)^3 + 6(j\omega)^2 + 8(j\omega) + k = 0$$

$$-j\omega^3 + (6)(-1)\omega^2 + 8j\omega + k = 0$$

$$-j\omega^3 - 6\omega^2 + j8\omega + k = 0$$

Separating real and imaginary parts:

$$-j\omega^3 + j8\omega = 0 \text{ (imag)}$$

$$-6\omega^2 + k = 0 \text{ (real)}$$

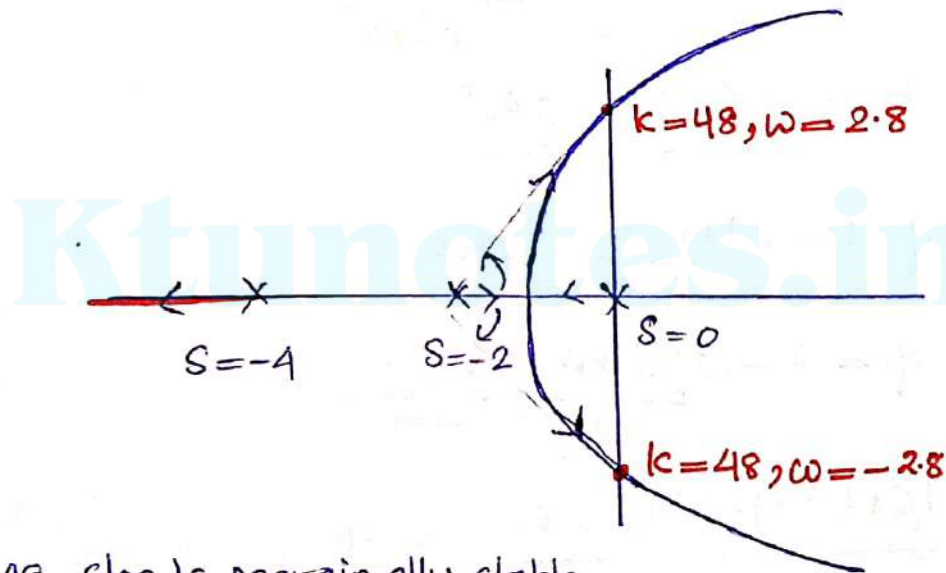
$$\omega^2 = 8$$

$$k = 6\omega^2 = \underline{\underline{48}}$$

$$\omega = \sqrt{8} = \underline{\underline{\pm 2.8}}$$

$$\omega = \underline{\underline{\pm 2.8}} \text{ and } k = 48$$

The root locus will cross the point where it intersects the imaginary axis where k value is 48 and ω value is ± 2.8 .



→ So At $k = 48$, s/m is marginally stable

→ frequency of oscillation, $\omega = 2.8 \text{ rad/sec}$

→ The range for which s/m remains stable: $0 < k < 48$
(poles lies on LHS $\Rightarrow$ stable)

$k = 48$: marginally stable

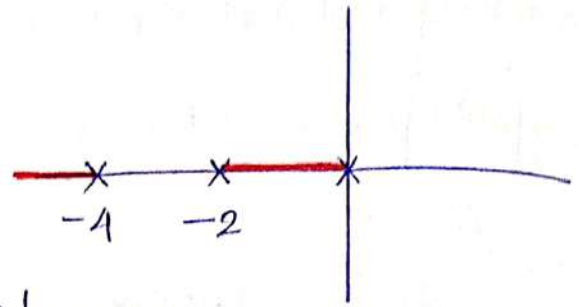
$k > 48$: poles shift to RHS of the plane
 $\Rightarrow$ unstable s/m

Qn) Sketch the root locus of the system whose open loop transfer fn is

$$G(s) = \frac{k}{s(s+2)(s+4)}$$

Poles are located at $s=0, -2, -4$

Root locus on real axis.



No zeros

3 root locus branches terminating at ∞ .

Asymptotes:

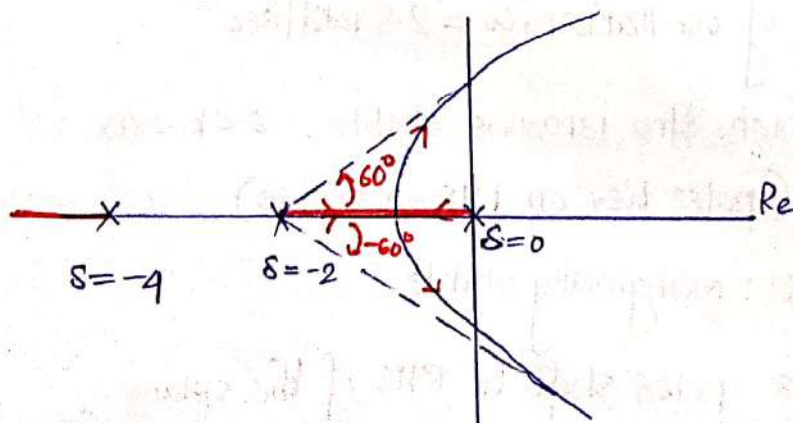
$$\text{angle of asymptotes} = \frac{\pm 180^\circ (2q+1)}{n-m}$$

$$q=0 \rightarrow \frac{\pm 180^\circ}{3} = \underline{\underline{\pm 60^\circ}}$$

$$q=1 \rightarrow \frac{\pm 180^\circ \times 3}{3} = \underline{\underline{\pm 180^\circ}}$$

$$q=2 \rightarrow \frac{\pm 180^\circ \times 5}{3} = \underline{\underline{\pm 300^\circ}}$$

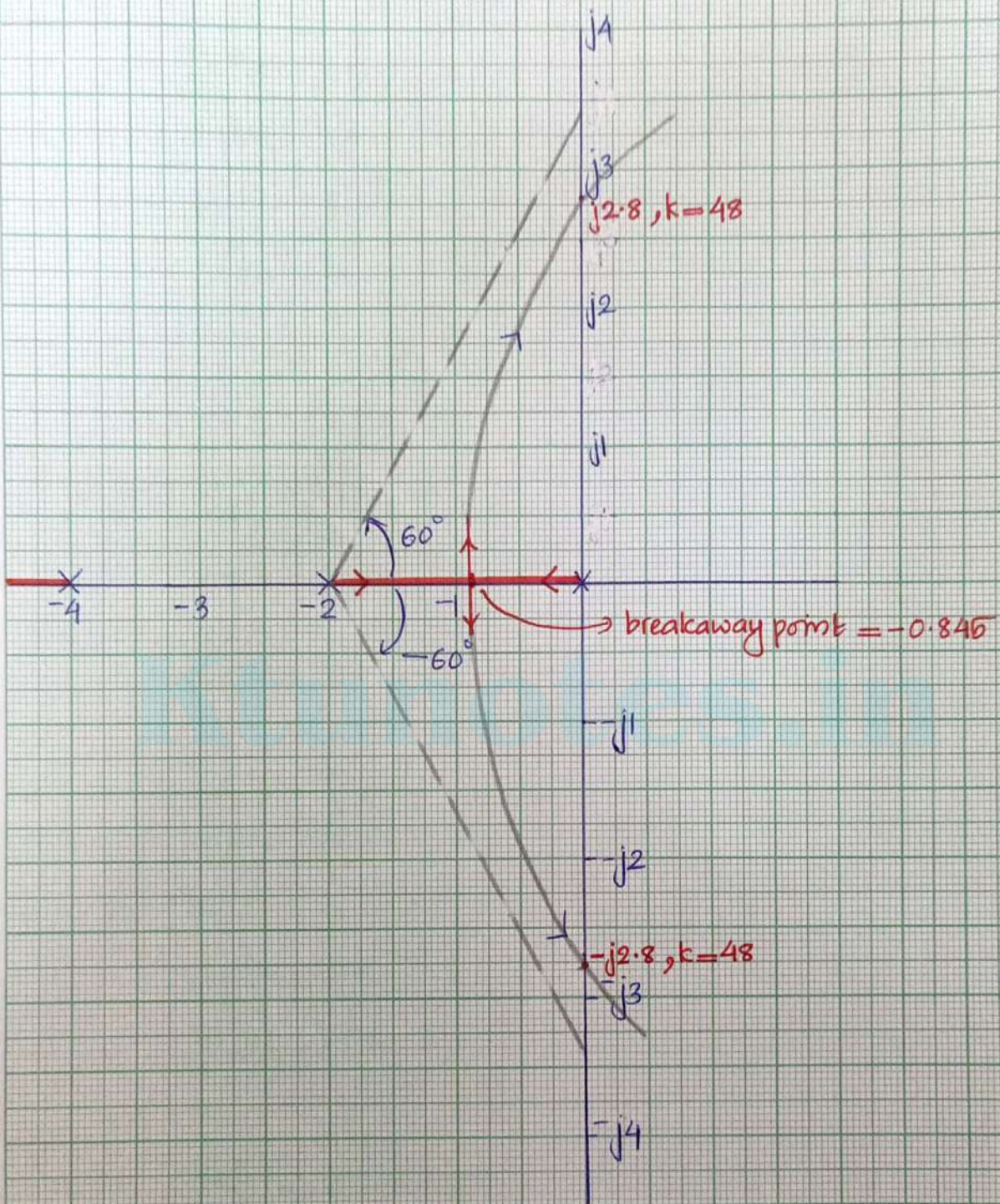
$$\text{Centroid} = \frac{(0 + (-2) + (-4)) - 0}{3} = -\frac{6}{3} = \underline{\underline{-2}}$$



Then find breakaway point

No complex poles/zeros $\Rightarrow$ no need to calculate angles of departure & arrival.

Scale
1 cm = 0.5 units



Then find ω and k

$$\omega = \pm 2.8, k = 48$$

Qn) Sketch the root locus of a unity feedback system whose open loop transfer function is $G(s) = \frac{k(s+9)}{s(s^2+4s+11)}$.

Poles $\rightarrow s=0, s=-2 \pm j2.64$

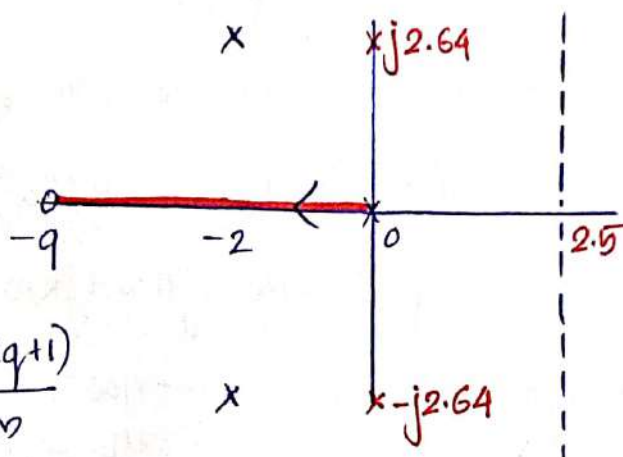
zeros $\rightarrow s=-9$

Angle of asymptotes $= \frac{\pm 180^\circ(2q+1)}{n-m}$

$n=3, m=1$

$q=0 \rightarrow \pm 90^\circ$

$q=1 \rightarrow \pm 270^\circ$



Centroid $= \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{[0 + (-2 + j2.64) + (-2 - j2.64)] - (-9)}{2}$

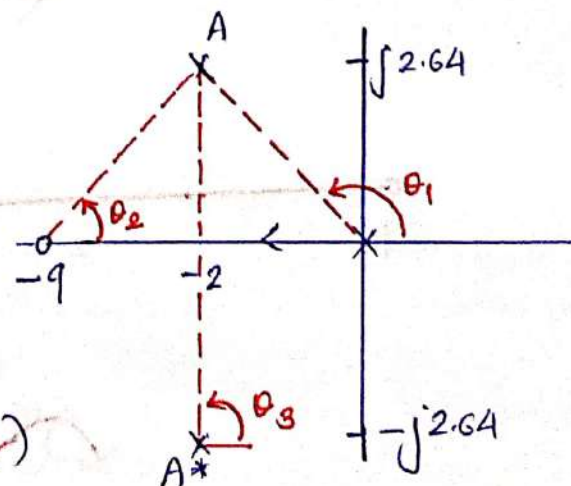
$$= \frac{-4 + 9}{2} = \frac{5}{2} = 2.5$$

Angle of departure:-

$\theta_1 = 180^\circ - \tan^{-1}\left(\frac{2}{2.64}\right) = 127.1^\circ$

$\theta_2 = \tan^{-1}\left(\frac{2.64}{1}\right) = 20.7^\circ$

$\theta_3 = 90^\circ$



Angle of departure $= 180^\circ - (\theta_1 + \theta_3) + (\theta_2)$
 (from complex pole A) $= 180^\circ - (127.1^\circ + 90^\circ) + 20.7^\circ$
 $= -16.4^\circ$

Angle of departure from complex pole $A^* = -16.4^\circ$
Crossing point on imaginary axis

$$G(s) = \frac{k(s+9)}{s(s^2+4s+11)} \quad H(s)=1$$

$$TF = \frac{G(s)}{1+G(s)}$$

Characteristic eqn $\Rightarrow s^3 + 4s^2 + 11s + ks + 9k = 0$

$$(j\omega)^3 + 4(j\omega)^2 + 11(j\omega) + k(j\omega) + 9k = 0$$

$$-j\omega^3 - 4\omega^2 + j11\omega + jk\omega + 9k = 0$$

$$-j\omega^3 + j11\omega + jk\omega = 0$$

$$\omega^3 = 11\omega + k\omega$$

$$\omega^2 = 11 + k \quad \text{--- ①}$$

$$-4\omega^2 + 9k = 0$$

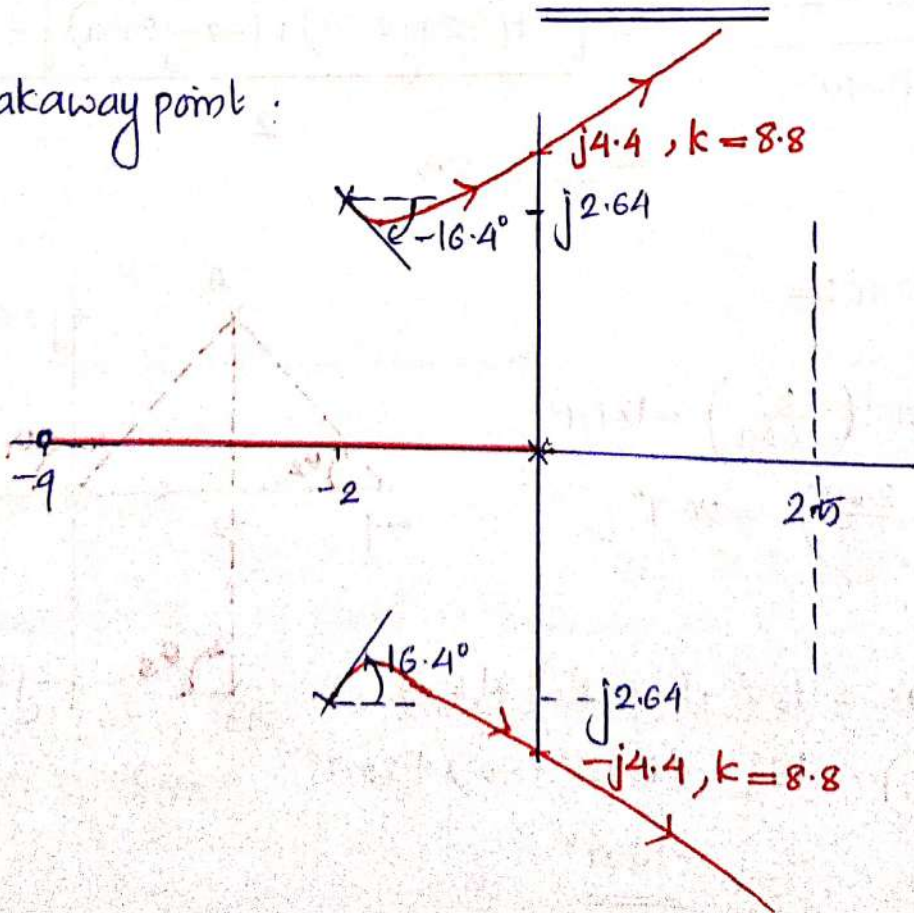
$$9k = 4\omega^2 \quad \text{--- ②}$$

$$\text{① in ② } 9k = 4(11+k)$$

$$k = 8.8$$

$$\omega = \pm 4.4$$

No breakaway point :



$0 < k < 8.8 \rightarrow \text{stable}$

$k = 8.8 \rightarrow \text{marginally stable}$

$k > 8.8 \rightarrow \text{unstable}$

Advantage of root locus: We're able to comment on the closed loop system stability based on the OLTF and from the corresponding root locus.

Qn) Sketch the root locus of a unity feedback system with open loop transfer function $G(s) \cdot H(s) = \frac{k(s+1.5)}{s(s+1)(s+5)}$.

3 poles and 1 zero

Poles $\Rightarrow s = 0, -1, -5$

Zeros $\Rightarrow s = -1.5$

Angle of asymptotes = $\frac{\pm 180^\circ (2q+1)}{n-m}$

$$n-m=2$$

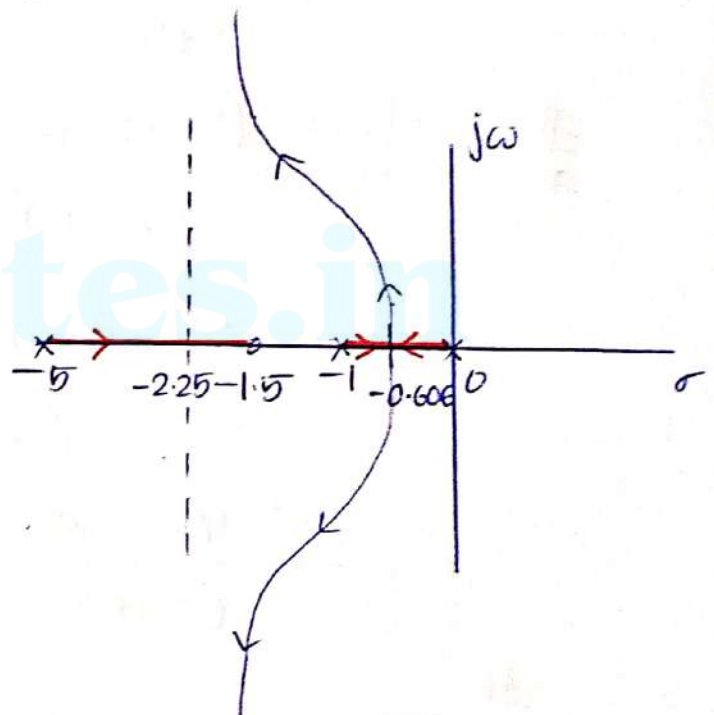
$$q=0$$

$$\frac{\pm 180^\circ}{2} = \pm 90^\circ$$

$$q=1$$

$$\frac{\pm 180^\circ (3)}{2} = \pm 270^\circ$$

$$\begin{aligned} \text{Centroid} &= \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{(0-1-5) - (-1.5)}{2} \\ &= \frac{-6+1.5}{2} = \underline{\underline{-2.25}} \end{aligned}$$



Breakaway point:

characteristic eqn: $G(s)H(s) = \frac{k(s+1.5)}{s(s+1)(s+5)}$

CLTF = $\frac{G(s)}{1+G(s)H(s)}$

$$s(s+1)(s+5) + k(s+1.5) = 0$$

$$s^3 + 6s^2 + 5s + k(s+1.5) = 0$$

$$k = -\frac{(s^3 + 6s^2 + 5s)}{s+1.5}$$

$$\frac{dk}{ds} = \frac{-(s+1.5)(3s^2+12s+5) - (-(s^3+6s^2+5s)(1))}{(s+1.5)^2}$$

$$= \frac{-3s^3 + 12s^2 - 5s + 4.5s^2 - 18s - 7.5 + s^3 + 6s^2 + 5s}{(s+1.5)^2}$$

$$= \frac{-2s^3 - 10.5s^2 - 18s - 7.5}{(s+1.5)^2}$$

$$\frac{dk}{ds} = 0$$

$$\Rightarrow \frac{-2s^3 - 10.5s^2 - 18s - 7.5}{(s+1.5)^2} = 0$$

$$s = -0.606, -2.32 \pm j0.89$$

$$s = -0.606$$

$$k = - \frac{[(-0.606)^3 + 6[-0.606]^2 + 5(-0.606)]}{-0.606 + 1.5}$$

$$= \underline{\underline{1.17}} \text{ +ve and real}$$

$\therefore -0.606$ is actual breakaway point

Crossing point of R-L branch on Imaginary axis:

$$s^3 + 6s^2 + 5s + ks + k(1.5) = 0$$

$$s = j\omega$$

$$(j\omega)^3 + 6(j\omega)^2 + 5(j\omega) + k(j\omega) + k(1.5) = 0$$

$$-j\omega^3 - 6\omega^2 + 5j\omega + kj\omega + 1.5k = 0$$

$$-6\omega^2 + 1.5k + j(-\omega^3 + 5\omega + k\omega) = 0$$

$$-6\omega^2 + 1.5k = 0 \quad \text{--- (1)}$$

$$-\omega^3 + 5\omega + k\omega = 0 \quad \text{--- (2)}$$

$$6\omega^2 = 1.5k$$

$$\omega^3 = 5\omega + k\omega$$

$$\omega^2 = (5+k)$$

$$6(5+k) = 1.5k$$

$$30 + 6k = 1.5k$$

$$\underline{\underline{k = -6.67}}$$

So ω is imaginary

$\therefore \omega^2 = 5 - 6.67$ which is -ve

The root locus is drawn for the values from 0 to ∞ . Here, it is -ve
So that indicates that it is not crossing the imaginary axis. It
→ It'll move towards asymptotes (follow asymptotic path to ∞)
 $k = 0 \text{ to } \infty \Rightarrow \text{stable}$

Sketch the root locus of a unity feedback s/m with open loop transfer function. $G(s)H(s) = \frac{k}{s(s+4)(s^2+4s+20)}$

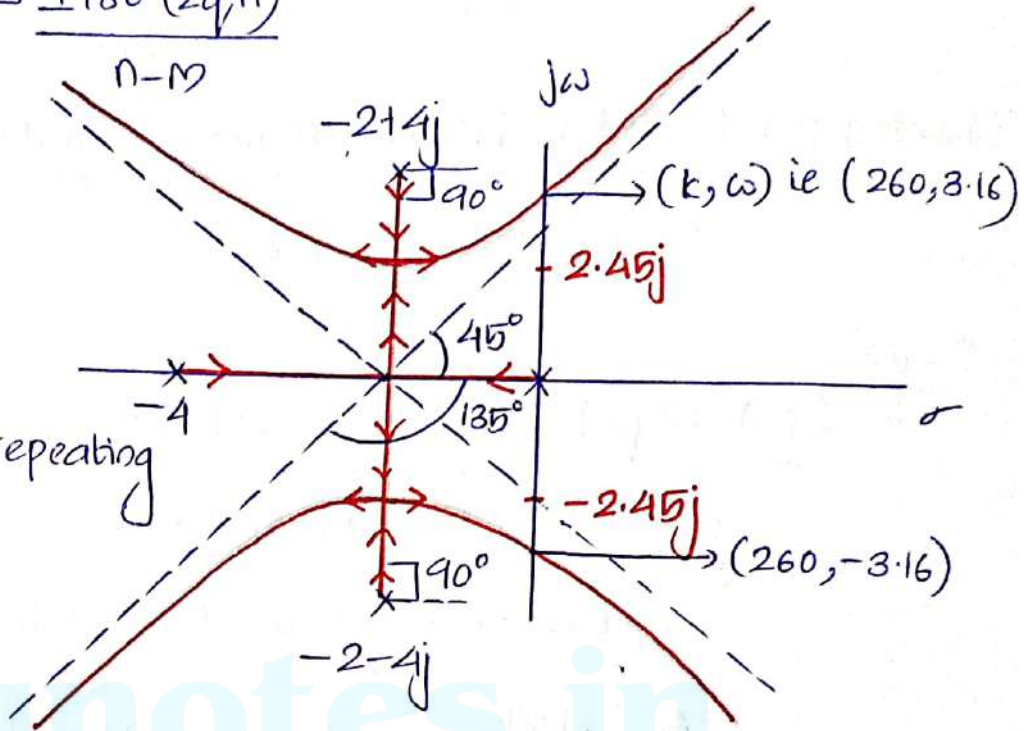
poles = $0, -4, -2 \pm 4j$

Angle of asymptotes, $\theta = \frac{\pm 180^\circ(2q+1)}{n-m}$

for $q=0$, $\theta = \frac{\pm 180}{4}$
 $= \pm 45^\circ$

for $q=1$, $\theta = \pm 135^\circ$

$q=2$, angles will be repeating
 $\theta = \pm 225^\circ$



Centroid = $\frac{(0 - 4 - 2 + 4j - 2 - 4j) - 0}{4} = -2$

Break away point:

$$s(s+4)(s^2+4s+20) + k = 0$$

$$s^4 + 8s^3 + 36s^2 + 80s + k = 0$$

$$k = -s^4 - 8s^3 - 36s^2 - 80s$$

$$\frac{dk}{ds} = -4s^3 - 24s^2 - 72s - 80 = 0$$

$$s = -2 \pm 2.45j, -2$$

$s = -2$; $k = -2^4 - 8(-2)^3 - 36 \times 4 - 80(-2) = \underline{64}$ Real and +ve

$s = -2$, break away point

$$\left. \begin{array}{l} s = -2 + 2.45j \quad k = 100 \\ s = -2 - 2.45j \quad k = 100 \end{array} \right\} \text{Also real and true}$$

The value of k is real and true for all roots which indicates ^{that} all three roots are valid breakaway points.

But, -2 is on the real axis.

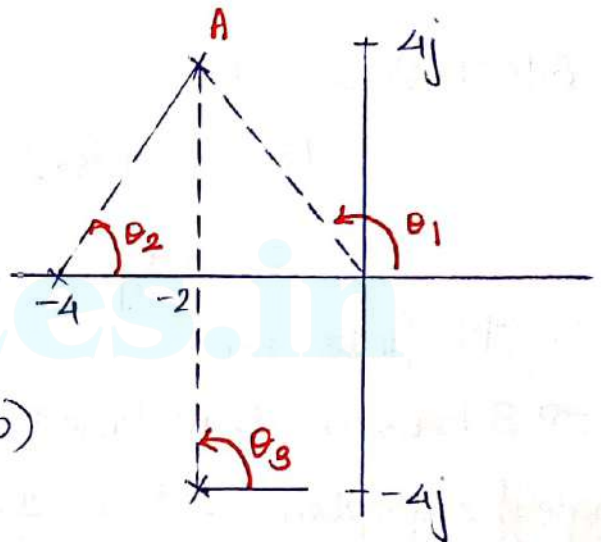
Angle of departure from A

$$= 180^\circ [\theta_1 + \theta_2 + \theta_3]$$

$$\theta_1 = 180^\circ - \tan^{-1} \frac{4}{2} = 117^\circ$$

$$\theta_2 = \tan^{-1} \frac{4}{2} = 63^\circ$$

$$\theta_3 = 90^\circ$$



$$\begin{aligned} \text{Angle of departure from A} &= 180^\circ - (117 + 63 + 90) \\ &= \underline{\underline{-90^\circ}} \end{aligned}$$

$$\text{from } A^* = \underline{\underline{90^\circ}}$$

Crossing point on imaginary axis

$$s^4 + 8s^3 + 36s^2 + 80s + k = 0$$

$$s = j\omega$$

$$\omega^4 - 8j\omega^3 - 36\omega^2 + 80j\omega + k = 0$$

$$\omega^4 - 36\omega^2 + k = 0$$

$$-8\omega^3 + 80\omega = 0$$

$$80\omega = 8\omega^3$$

$$8\omega^2 = 80$$

$$\omega^2 = 10$$

$$\omega = \pm 3.162 \text{ rad/sec}$$

$$k = 36\omega^2 - \omega^4$$

$$= 360 - 100$$

$$= \underline{\underline{260}}$$

$0 < k < 260$ - stable

$k > 260$ - unstable

HW problem

$$Qn: G(s)H(s) = \frac{k}{(s+2)(s^2+2s+2)}$$

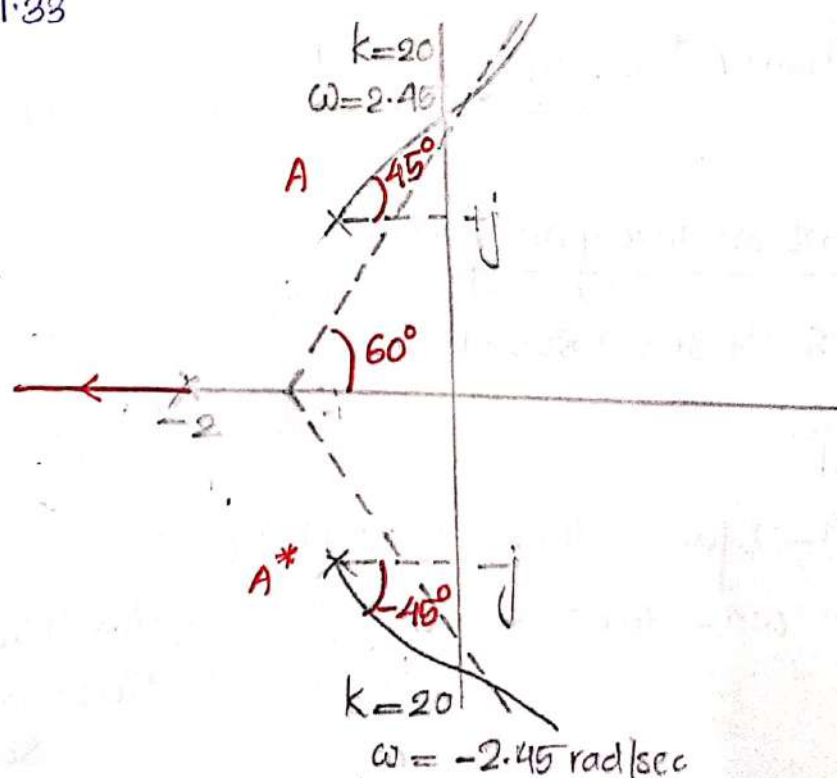
$$s = -2, -1 \pm j$$

No finite zeros

→ 3 locus root locus branches going to ∞ .

Angle of asymptotes = $\pm 60^\circ, \pm 180^\circ$

Centroid = -1.33



Angle of departure from A = 45°
 from A* = -45°

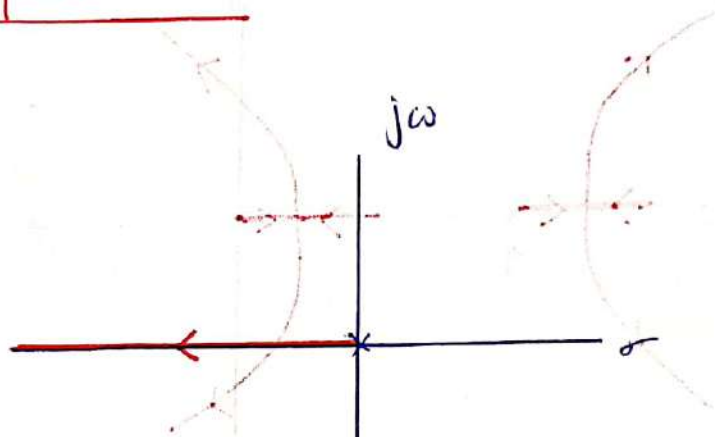
The crossing point on imaginary axis :

$$k = 20$$

$$\omega = \pm 2.45 \text{ rad/sec}$$

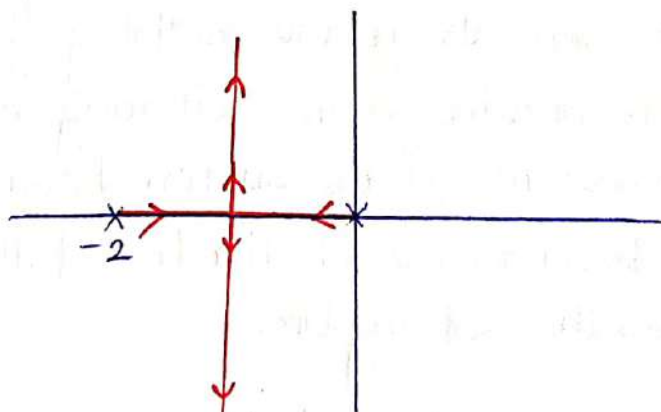
Effect of adding poles in root locus

$$\textcircled{1} G(s) \cdot H(s) = \frac{k}{s}$$



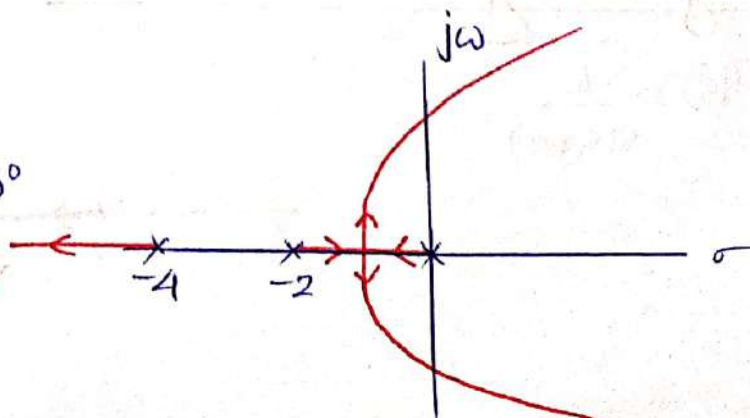
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$$\textcircled{2} G(s)H(s) = \frac{k}{s(s+2)}$$



$$\textcircled{3} G(s) \cdot H(s) = \frac{k}{s(s+2)(s+4)}$$

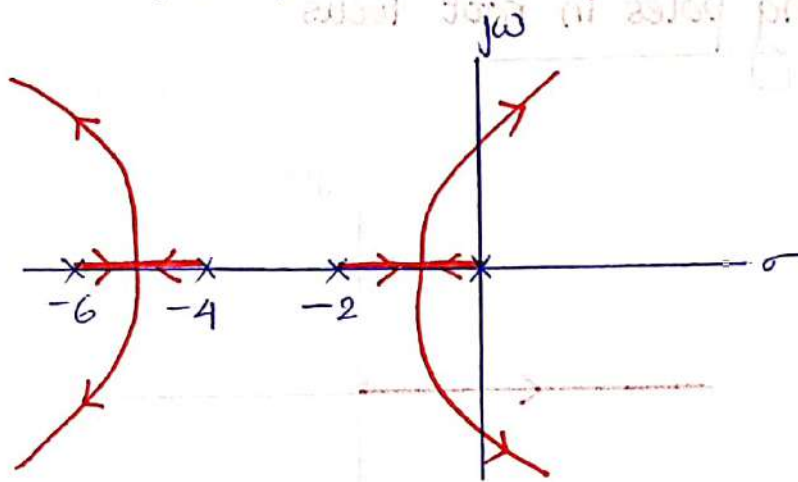
Angle of asymptote = $\pm 60^\circ$



Note: When we keep on adding poles to the s/m (on to the LHS), the root locus branch moves to the right half of s-plane, thus lowering the relative stability of the s/m.

Stability of s/m ↓ as poles are added to the s/m.

④ $G(s) \cdot H(s) = \frac{k}{s(s+2)(s+4)(s+6)}$

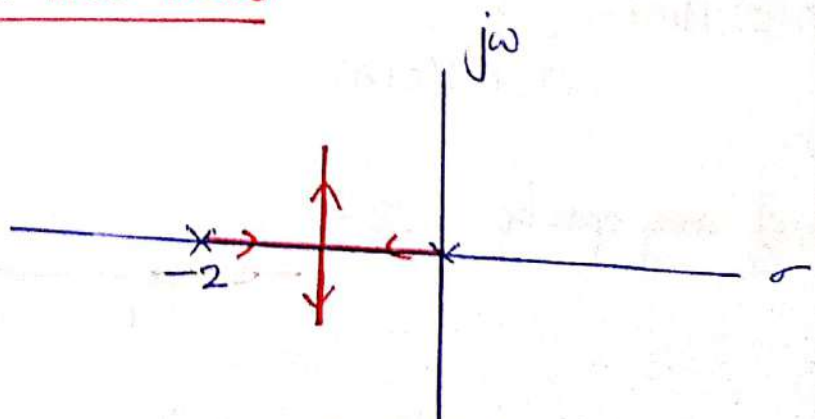


☆ Effect of adding poles in root locus:

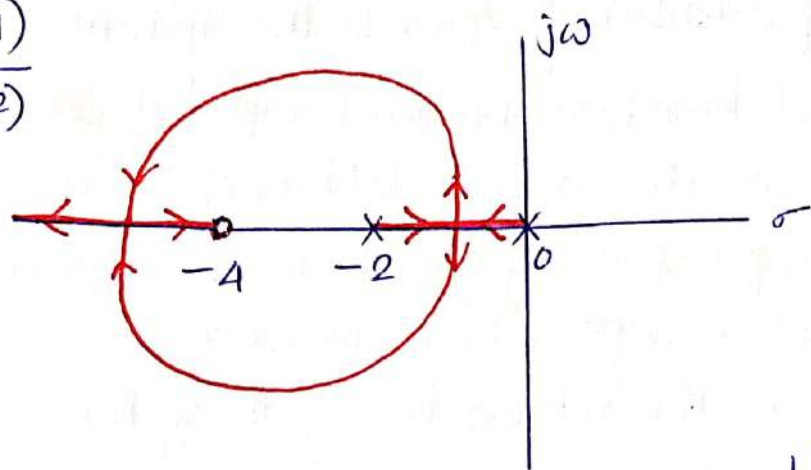
- Root locus branches moves towards right half of s-plane
- Reduces / lowers the relative stability of the system.
- The system becomes more oscillatory in nature.
- The step response of the system becomes slower.
- Since s/m becomes slower, rise time, fall time increases.
- It increases the settling time.

Effect of adding zeros in root locus

① $G(s) \cdot H(s) = \frac{k}{s(s+2)}$

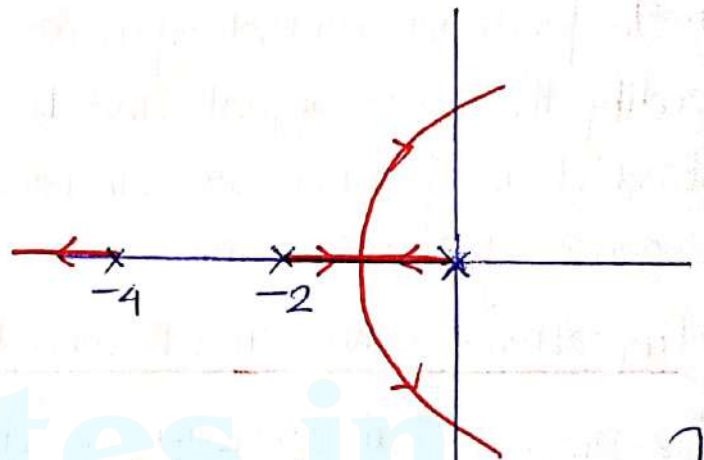


$$(2) G(s) \cdot H(s) = \frac{k(s+4)}{s(s+2)}$$



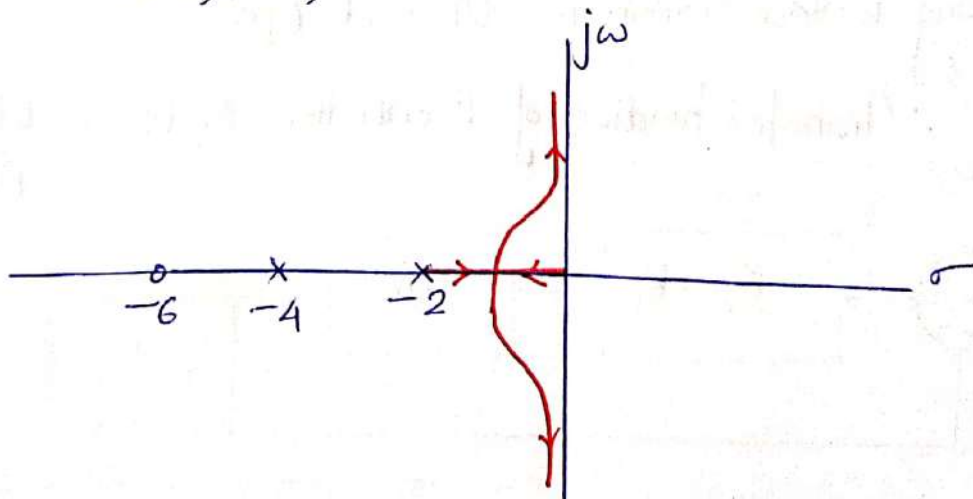
On adding more zeros, it will pull root locus towards left side.
So, stability $\uparrow$.

$$(3) G(s) \cdot H(s) = \frac{k}{s(s+2)(s+4)}$$



{ On adding more zeros, it will pull root locus towards left side, }
So stability $\uparrow$ s.

$$(4) G(s) \cdot H(s) = \frac{k(s+6)}{s(s+2)(s+4)}$$



* Effect of addition of zeros in the system

- The root locus branch shifts leftwards
- Improves the relative stability of the s/m.
- The step response of the s/m becomes faster.
- Rise time and peak time decreases.
- It reduces the settling time of the system.

P, PI, PD AND PID CONTROLLERS

→ In feedback control systems, a controller may be introduced to modify the error signal and to achieve a better control action. The introduction of controller will modify the transient and steady state response of the system.

1) Proportional Controllers (P-controller)

The proportional controller is a device that produces an output signal, $u(t)$ which is proportional to the error signal $e(t)$.

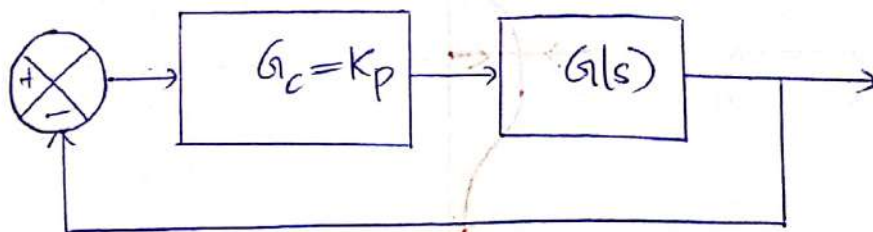
$$\text{ie } u(t) \propto e(t)$$

$$\therefore u(t) = K_p e(t)$$

K_p - proportional gain or constant

Taking Laplace transform $U(s) = K_p E(s)$

$$\therefore \text{Transfer Function of P controller } G_c(s) = \frac{U(s)}{E(s)} = K_p$$



The proportional controller improves the:

1. Steady state tracking accuracy.
2. Disturbance signal rejection.
3. Relative stability.

→ The drawback of proportional controller is that, it produces a constant steady state error.

So, in order to overcome this drawback, another type of controller known as proportional plus integral controller can be used.

2) Proportional plus Integral controller (PI controller)

PI controller is a device that produces an o/p signal $u(t)$ consisting of two terms — one proportional to input signal, $e(t)$ and the other proportional to the integral of the i/p signal, $e(t)$.

$$\text{i.e., } u(t) \propto [e(t) + \int e(t) dt]$$

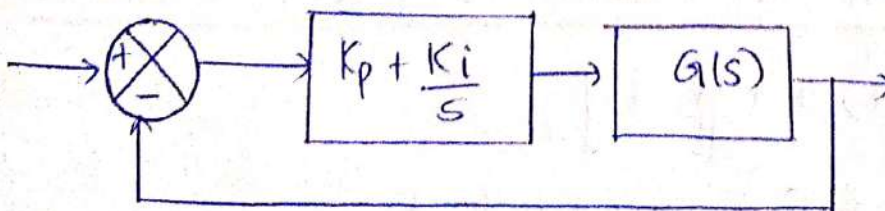
$$\therefore u(t) = k_p e(t) + k_i \int e(t) dt$$

k_p — proportional gain

k_i — Integral constant or gain.

Taking Laplace transform $U(s) = k_p E(s) + k_i \frac{E(s)}{s}$

$$\therefore \text{Transfer fn of PI controller, } G_c(s) = \frac{U(s)}{E(s)} = k_p + \frac{k_i}{s}$$



- PI controller reduces the steady state error, since the type no. of the system is increased by one. (as type no ↑ steady state error ↓)
- PI controller also increases the order of the system by one. The increase in the order of the system results in a less stable s/m

3) Proportional plus derivative controller (PD controller)

Output signal, $u(t)$ is proportional to error signal $e(t)$ and the derivative of error signal.

$$u(t) \propto \left[e(t) + \frac{d}{dt} e(t) \right]$$

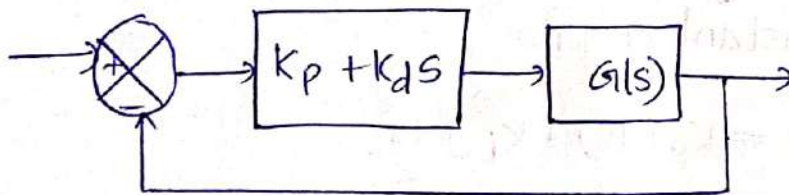
$$\therefore u(t) = K_p e(t) + K_d \frac{d}{dt} e(t)$$

K_p — Proportional gain

K_d — Derivative constant or gain.

Taking Laplace transform $U(s) = K_p E(s) + K_d s \cdot E(s)$

$$\therefore \text{Transfer Function of PD Controller, } G_c(s) = \frac{U(s)}{E(s)} = K_p + K_d s$$



PD controller introduces a zero in the s/m and it increases the damping ratio. This results in the reduction of peak overshoot.

4) Proportional plus Integral plus derivative controller (PID controller)

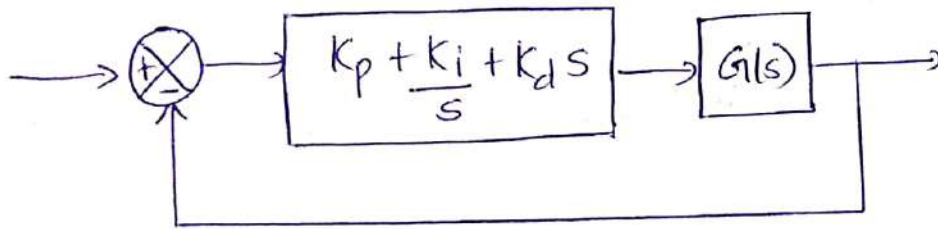
$$u(t) \propto \left[e(t) + \int e(t) dt + \frac{d}{dt} e(t) \right]$$

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

Taking Laplace transform $U(s) = K_p E(s) + K_i \frac{E(s)}{s} + K_d s E(s)$

$\therefore$ Transfer fn of PID controller, $G_c(s) = \frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s$

- The PID controller have the combined effect of all the three controllers – Proportional, derivative and integral control actions.
- Hence the introduction of PID controller stabilises the gain, reduces the steady state error and peak overshoot of the sys.



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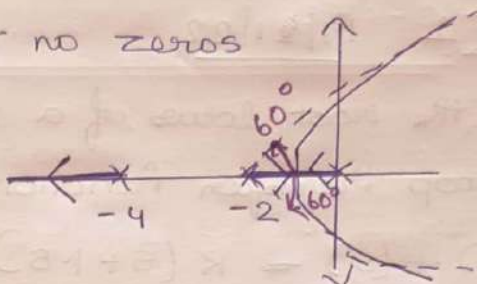
Q. Sketch the root locus of the S/m whose open loop transfer function is $G(s) = \frac{K}{s(s+2)(s+4)}$

① Poles are located on $s = 0, -2, -4$

3 root locus branches

will terminate @ ∞ $\therefore$ no zeros

② Root locus on real axis



③ Asymptotes $\pm \frac{180^\circ}{3} = \pm 60^\circ$ $q=0$

$$q=1 \Rightarrow \pm \frac{180 \times 3}{3} = \pm 180^\circ$$

$$q=2 \Rightarrow \pm \frac{180 \times 5}{3} = \pm 300^\circ$$

$n = \text{no. of poles} = 3$

$m = \text{no. of zeros} = 0$

$(n-m) = 3$ root locus

branches will get terminated @ ∞

No. of asymptotes $= n-m$
 $= 3$

$$\text{Centroid} = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$$

$$= \frac{(0 + (-2) + (-4)) - 0}{3}$$

$$= \frac{-6}{3} = \underline{\underline{-2}}$$

We have already calculated the break away point for this question i.e., $s = -0.845$

No complex poles / zeros so no need to find angle of departure & angle of arrival

→ Crossing point on imaginary axis = ± 2.8
 $K = 48$

Live - 1 17/11/22

Q) Sketch the root locus of a unity feedback s/m with open loop transfer functions

$$G(s) \cdot H(s) = \frac{K(s+1.5)}{s(s+1)(s+5)}$$

3 poles and 1 zero

So 3 root locus

branches, 1 will terminate @ the 1 zero and rest of the root locus will go to ∞ .

$$\text{Poles} = s = 0, -1, -5$$

$$\text{Zeros} = s = -1.5$$

→ angle of asymptotes = $\pm \frac{180^\circ(2q+1)}{n-m}$

$$q = 0, \pm 90^\circ$$

$$q = 1, \pm 270^\circ$$

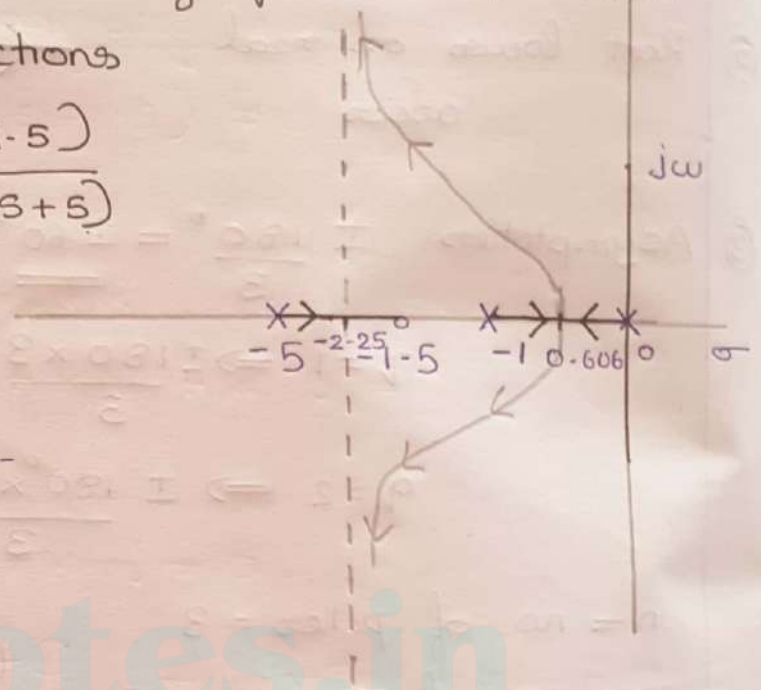
$$\frac{n-m}{\text{poles} \quad \text{zeros}}$$

$$3-1=2$$

→ Centroid = $\frac{\sum \text{poles} - \sum \text{zeros}}{n-m}$

$$= \frac{(0 + (-1) + (-5) - (-1.5))}{2}$$

$$= \frac{-6 + 1.5}{2} = \underline{\underline{-2.25}}$$



Break away point

1) characteristic eqn

$$G(s) \cdot H(s) = \frac{K(s+1.5)}{s(s+1)(s+5)}$$

$$CLTF = \frac{G(s)}{1 + G(s) \cdot H(s)}$$

$$= s(s+1)(s+5) + K(s+1.5) = 0$$

$$= s^3 + 6s^2 + 5s + K(s+1.5) = 0$$

$$K = - \frac{(s^3 + 6s^2 + 5s)}{(s+1.5)}$$

$$\frac{dK}{ds} = 0$$

$$\frac{dK}{ds} = \frac{-(s+1.5)(3s^2 + 12s + 5) - [-(s^3 + 6s^2 + 5s)]}{(s+1.5)^2}$$

$$= \frac{-3s^2 - 4.5s^2 - 12s^2 - 18s - 5s - 7.5 + s^3 + 6s^2 + 5s}{(s+1.5)^2}$$

$$= \frac{-2s^3 - 10.5s^2 - 18s - 7.5}{(s+1.5)^2}$$

$$= 2s^3 + 10.5s^2 + 18s + 7.5 = 0$$

$$s = \underline{\underline{-0.606, -2.32 \pm j0.89}}$$

Break away pt b/w 0 & -1 $\therefore -0.606$ OR.

$s = -0.606$ is +ve & real - actual break away point.

$$K = - \left[\frac{(-0.606)^3 + 6(-0.606)^2 + 5(-0.606)}{-0.606 + 1.5} \right]$$

$$= 1.17 \quad \text{+ve \& real}$$

$\therefore -0.606$ is actual break away pt

Crossing pt on imaginary axis

$$s^3 + 6s^2 + 5s + K(s + 1.5) = 0$$

$$s = j\omega$$

$$(j\omega)^3 + 6(j\omega)^2 + 5(j\omega) + K(j\omega + 1.5) = 0$$

$$-j\omega^3 - 6\omega^2 + 5j\omega + Kj\omega + K1.5 = 0$$

$$-6\omega^2 + 1.5K = 0$$

$$+j(-\omega^3 + 5\omega + K\omega) = 0$$

$$\therefore K = \frac{6\omega^2}{1.5} \quad \text{--- (1)}$$

$$-\omega^3 + 5\omega + K\omega = 0 \quad \text{--- (2)}$$

$$\omega^3 = 5\omega + K\omega$$

$$\omega^2 = 5 + K$$

$$K = \omega^2 - 5$$

$$\omega^2 - 5 = \frac{6\omega^2}{1.5}$$

$$1.5\omega^2 - 7.5 = 6\omega^2$$

$$1.5\omega^2 - 6\omega^2 = 7.5$$

$$\omega^2 (-4.5) = 7.5$$

$$\omega^2 = -1.666$$

$$\omega =$$

$$K = -6.67$$

↓ root locus branch

Indicates $\wedge$ not crossing imaginary axis

So, it will move towards asymptotes

Follow asymptotic path to ∞

→ No complex poles/zeros so no need of angle of departure / angle of arrival

* s/m stable

LIVE - 2 18/1/22

Q) Sketch the root locus of a unity feedback s/m with OLTF

$$G(s) \cdot H(s) = \frac{K}{s(s+4)(s^2+4s+20)}$$

$$\text{Poles} = 0, -4, -2 \pm 4j$$

angle of asymptotes

$$\theta = \pm \frac{180^\circ(2q+1)}{n-m}$$

$$= \pm \frac{180}{4} \quad q=0$$

$$= \pm 45^\circ$$

$$q=1, \theta = \pm 135^\circ$$

$q=2$, angles will be repeating

$$\theta = \pm 225^\circ$$

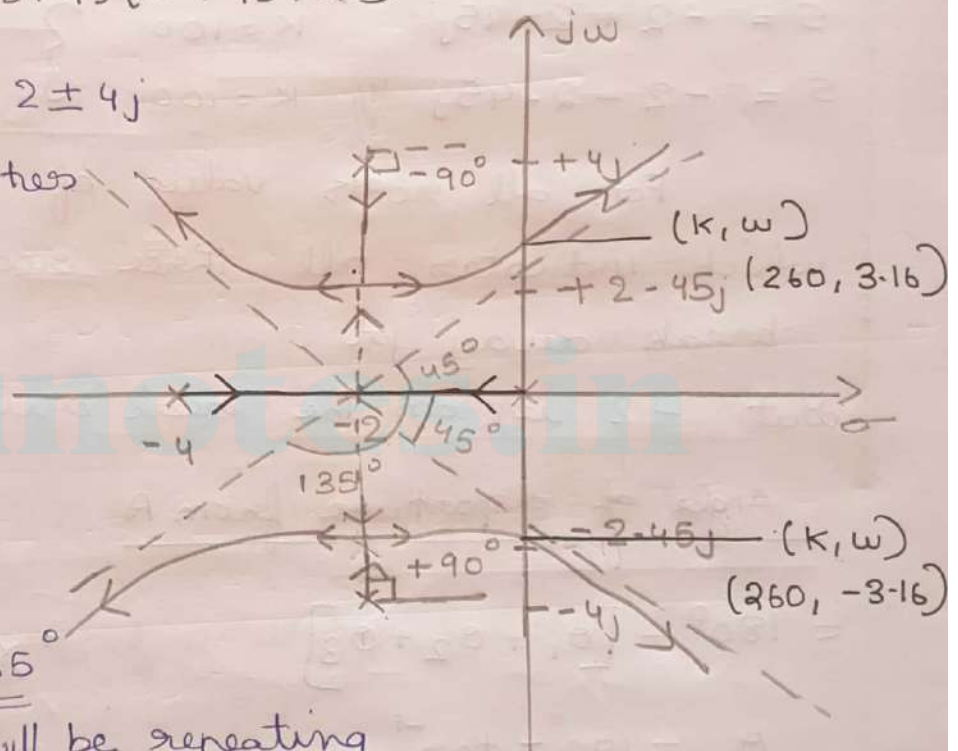
$$\text{centroid} = \frac{(0 - 4 - 2 + 4j - 2 - 4j) - 0}{4}$$

$$= -2$$

Break away point

$$s(s+4)(s^2+4s+20) + K = 0$$

$$s^4 + 8s^3 + 36s^2 + 80s + K = 0$$



$$K = -s^4 - 8s^3 - 36s^2 - 80s$$

$$\frac{dK}{ds} = -4s^3 - 24s^2 - 72s - 80 = 0$$

$$s = -2 \pm 2.45j, -2$$

$$s = -2$$

$$K = -2^4 - 8(-2)^3 - 36 \times 4 - 80(-2)$$

$$= 64$$

Real & +ve

$s = -2$, break away pt

$$\begin{array}{ll} s = -2 + 2.45j & K = 100 \\ s = -2 - 2.45j & K = 100 \end{array} \quad \left. \vphantom{\begin{array}{l} s = -2 + 2.45j \\ s = -2 - 2.45j \end{array}} \right\} \text{Also real \& +ve}$$

For all roots value of K +ve & real.
which indicates all three roots are valid
break away pt

But -2 is on the real axis

Angle of departure from A

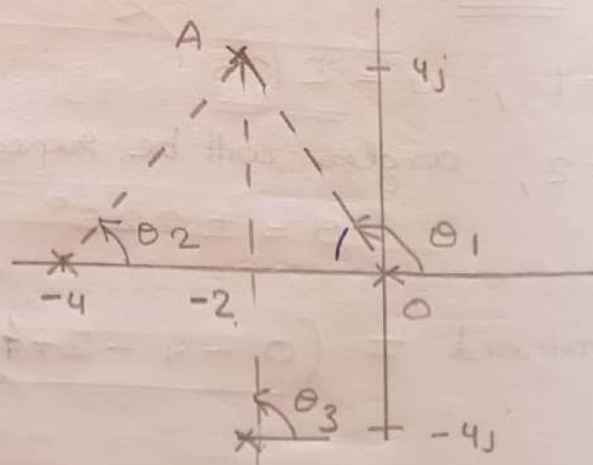
$$= 180^\circ - [\theta_1 + \theta_2 + \theta_3]$$

$$\theta_1 = 180^\circ - \tan^{-1} \frac{4}{2}$$

$$= 117^\circ$$

$$\theta_2 = \tan^{-1} \frac{4}{2} = 63^\circ$$

$$\theta_3 = 90^\circ$$



$$\begin{array}{l} \text{angle of departure} = 180^\circ - (117 + 63 + 90) \\ \text{from A} \quad \quad \quad = -90^\circ \end{array}$$

$$\text{from } A^* = 90^\circ$$

Crossing point on imaginary axis

$$s^4 + 8s^3 + 36s^2 + 80s + K = 0$$

$$s = j\omega$$

$$\omega^4 - 8j\omega^3 - 36\omega^2 + 80j\omega + K = 0$$

$$\omega^4 - 36\omega^2 + K = 0$$

$$-8\omega^3 + 80\omega = 0$$

$$80\omega = 8\omega^3$$

$$80 = 8\omega^2$$

$$\omega^2 = 10$$

$$\omega = \pm 3.162 \text{ rad/sec}$$

$$K = 36\omega^2 - \omega^4$$

$$= 36 \times 10 - 100$$

$$= 260$$

$0 < K < 260$ - Stable

$K > 260$ - unstable

H.I.W

$$K = 20$$

$$\omega = 2.45 \text{ rad/sec}$$

$$s = -2, -1 \pm j$$

$$\text{angle} = \pm 60^\circ, \pm 180^\circ$$

$$\text{centroid} = -1.33$$

$$\text{Angle of departure} = 45^\circ \text{ from A}$$

$$G(s)H(s) = \frac{K}{(s+2)(s^2+2s+2)}$$

$$\text{from } A^* = -45^\circ$$

Q) Sketch the root locus of a unity feedback S/m whose

$$\text{O.L.T.F is } G(s) = \frac{K(s+9)}{s(s^2+4s+11)}$$

$$\text{Poles} \rightarrow s = 0,$$

$$s = -2 \pm 2.64j$$

3 root locus

branches -

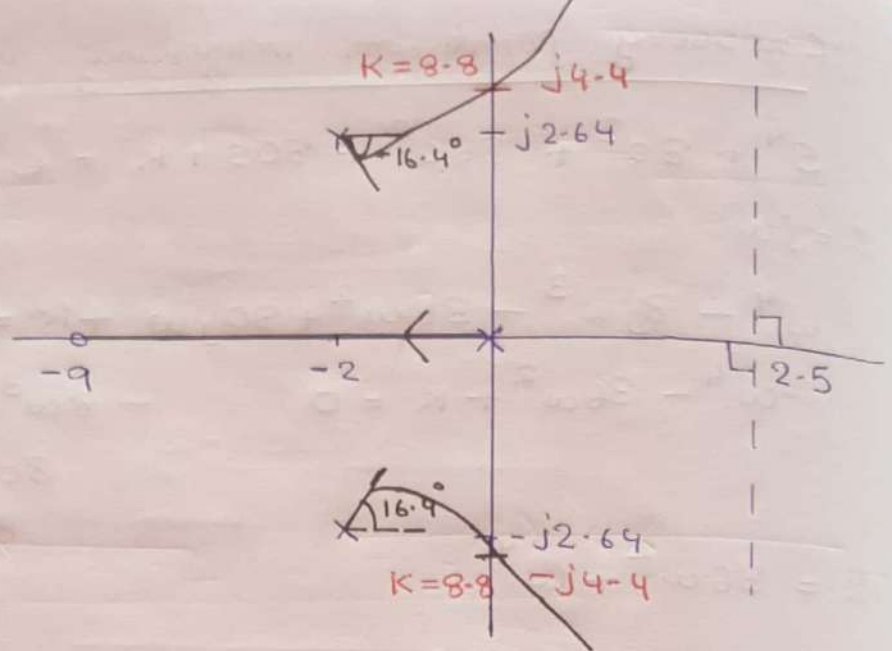
$$\text{Zeros} \rightarrow s = -9$$

Angle of asymptotes & centroid

$$\text{Angle of asymptotes} = \pm 180 \frac{(2q+1)}{n-m}$$

$$n = 3, m = +1$$

1 get terminated on zero in the real axis.
other two gets terminated at the zeroes @ ∞ .



$$q = 0 \rightarrow \pm 90^\circ$$

$$q = 1 \rightarrow \pm 270^\circ$$

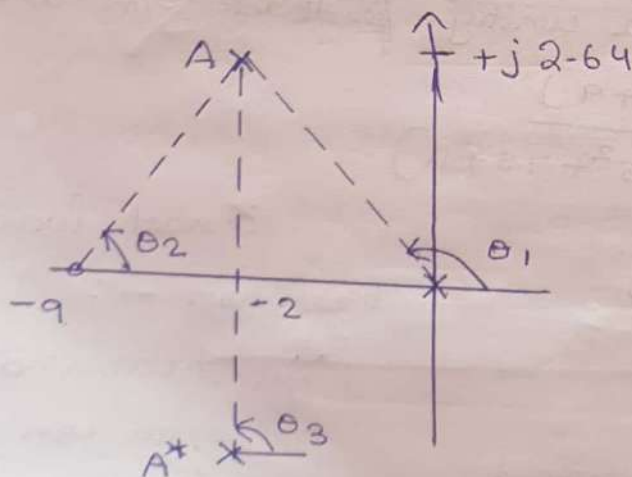
$270^\circ = \text{multiple of } 90^\circ$

So same

For $(q=2)$ all other

values of q , θ is a multiple of $\pm 90^\circ$.

Angle of departure.



$$\begin{aligned} \theta_1 &= 180^\circ - \tan^{-1} \frac{2}{2.64} \\ &= \underline{127.1^\circ} \end{aligned}$$

$$\begin{aligned} \theta_2 &= \tan^{-1} \frac{2.64}{7} \\ &= \underline{20.7^\circ} \end{aligned}$$

$$\theta_3 = \underline{90^\circ}$$

$$\begin{aligned} \text{Angle of departure from A} &= 180^\circ - (\theta_1 + \theta_3) + \theta_2 = 180^\circ - (127.1^\circ + 90^\circ) + 20.7^\circ \\ &= \underline{-16.4^\circ} \end{aligned}$$

Angle of departure from $A^* = \underline{\underline{16.4^\circ}}$

Crossing point on imaginary axis

$$G(s) = \frac{K(s+9)}{s(s^2+4s+11)}$$

$$|H(s)| = 1$$

$$T-F = \frac{G(s)}{1+G(s)}$$

$$CE \Rightarrow s^3 + 4s^2 + 11s + Ks + 9K = 0$$

$$s = j\omega \Rightarrow (j\omega)^3 + 4(j\omega)^2 + 11j\omega + Kj\omega + 9K = 0$$

$$-j\omega^3 - 4\omega^2 + j11\omega + jK\omega + 9K = 0$$

$$-j\omega^3 + j11\omega + jK\omega = 0$$

$$-4\omega^2 + 9K = 0$$

$$9K = 4\omega^2 \quad \text{--- (2)}$$

$$\omega^3 = 11\omega + K\omega$$

$$\omega^2 = 11 + K \quad \text{--- (1)}$$

① in ②

$$9K = 44 + 4K$$

$$\omega = \underline{\underline{\pm 4.4}}$$

$$5K = 44$$

$$K = \frac{44}{5} = \underline{\underline{8.8}}$$

No break away point since only one zero & one pole.

$0 < K < 8.8 \rightarrow$ stable (poles @ left)

$K = 8.8 \rightarrow$ marginally stable (poles @ imaginary axis)

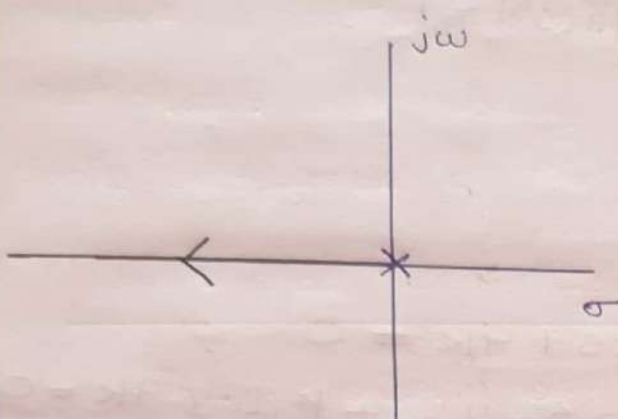
$K > 8.8 \rightarrow$ Roots @ right

$\therefore$ Unstable.

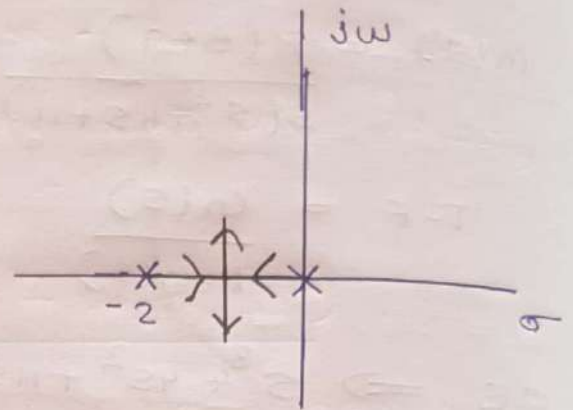
Effect of adding poles in root locus

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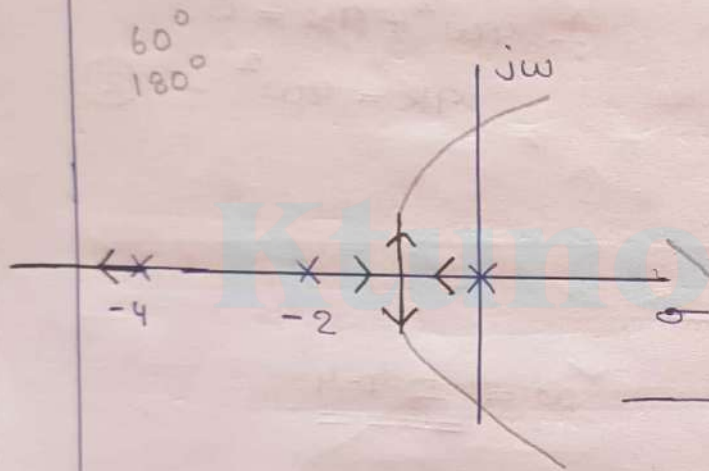
① $G(s) \cdot H(s) = \frac{K}{s}$



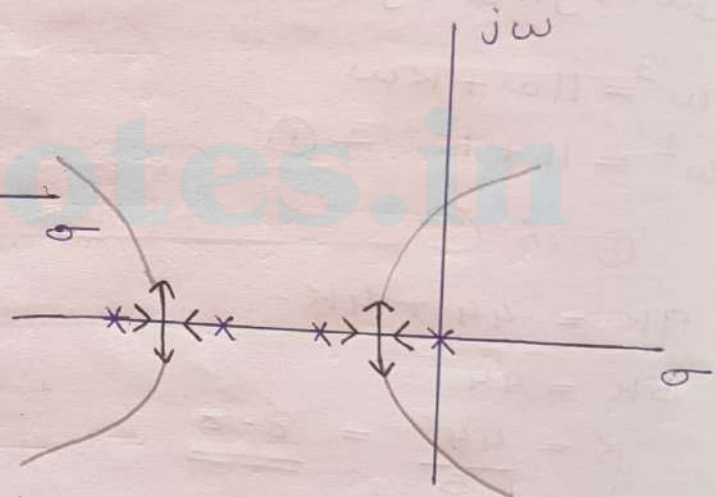
b) $G(s) \cdot H(s) = \frac{K}{s(s+2)}$



c) $G(s) \cdot H(s) = \frac{K}{s(s+2)(s+4)}$



d) $G(s) \cdot H(s) = \frac{K}{s(s+2)(s+4)(s+6)}$



* On adding more poles, the root locus has a tendency to move towards right side and hence S/m becomes unstable.

poles $\uparrow$ stability $\downarrow$.

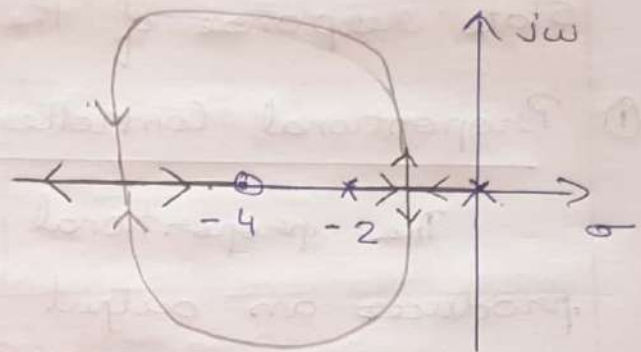
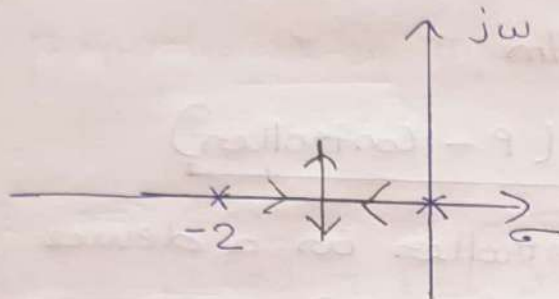
* $\therefore$ root locus branch is crossing the imaginary axis — S/m becomes more oscillatory
— Step response of the S/m becomes slower.
— $\uparrow$ the rise time, fall time.

— $\uparrow$ the settling time

Effect of adding zeros in root locus

$$G(s) \cdot H(s) = \frac{K}{s(s+2)}$$

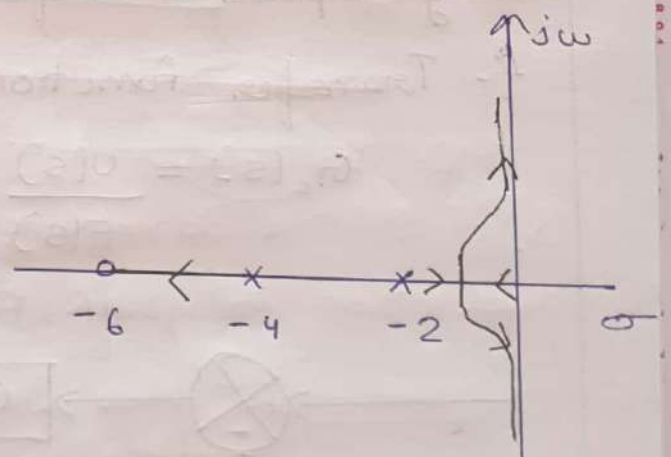
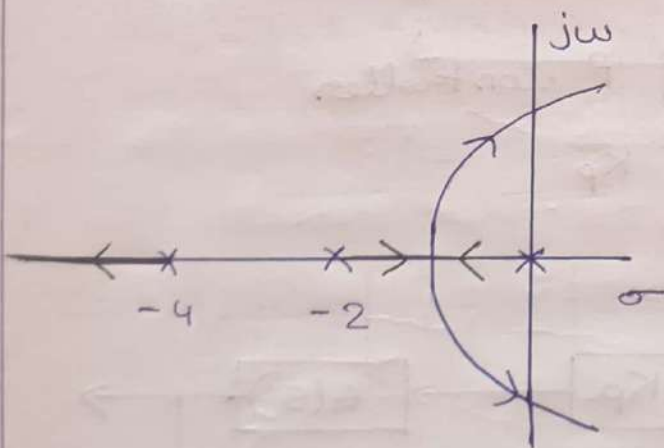
$$\frac{K(s+4)}{s(s+2)}$$



On adding more zeros, it will pull root locus towards left side so stability $\uparrow$.

c) $G(s) \cdot H(s) = \frac{K}{s(s+2)(s+4)}$

(d) $G(s) \cdot H(s) = \frac{K(s+6)}{s(s+2)(s+4)}$



- * $\uparrow$ stability of the s/m
- * Step response of the s/m faster.
- * Rise time, peak time $\downarrow$
- * Settling time of the s/m $\downarrow$.

P, PI, PD, PID Controllers

In feedback control systems, a controller may be introduced to modify the error signal and to achieve a better control action. The introduction of controllers will modify the transient and steady state response of the s/m.

① Proportional Controller (P-Controller)

The proportional controller is a device that produces an output signal, $u(t)$ which is proportional to the error signal $e(t)$

$$\text{i.e. } u(t) \propto e(t)$$

i/p - error signal

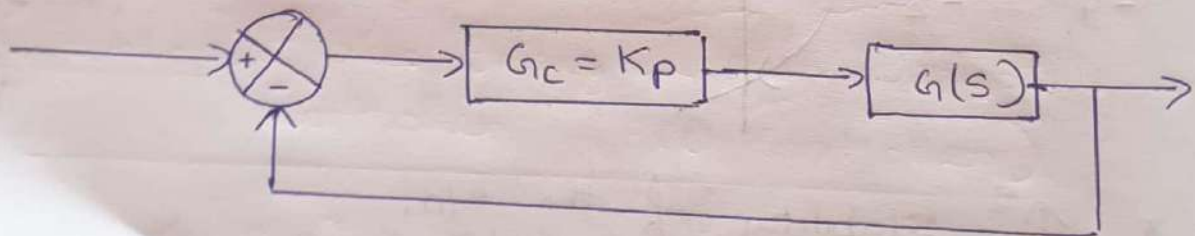
$$\therefore u(t) = K_p e(t)$$

K_p - proportional gain / constant

Taking Laplace transform $U(s) = K_p E(s)$

$\therefore$ Transfer function of P controller

$$G_c(s) = \frac{U(s)}{E(s)} = K_p$$



The proportional controller improves the

1. Steady state tracking accuracy
2. Disturbance signal rejection
3. Relative stability.

* The drawback of proportional controller is that ; it produces a constant steady state error.

② Proportional plus Integral controller (PI Controller)

PI controller is a device that produces an output signal, $u(t)$ consisting of two terms - one proportional to input signal, $e(t)$ and the other proportional to the integral of the input signal $e(t)$ i.e.,

$$u(t) \propto [e(t) + \int e(t) dt]$$

$$\therefore u(t) = K_p e(t) + K_i \int e(t) dt$$

K_p - proportional gain

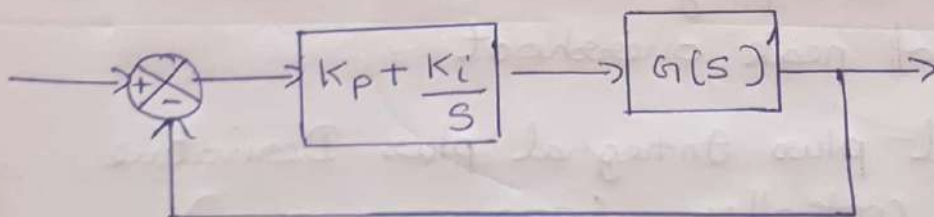
K_i - Integral constant or gain

Taking Laplace transform ;

$$U(s) = K_p E(s) + K_i \frac{E(s)}{s}$$

$\therefore$ Transfer function of PI controller ,

$$G_c(s) = \frac{U(s)}{E(s)} = K_p + \frac{K_i}{s}$$



P.I controller reduces the steady state error, since the type no: of the s/m is $\uparrow$ ed by one.

* PI controller also increases the order of the s/m by one. The $\uparrow$ in the order of the s/m results in a less stable s/m.

3. Proportional plus Derivative controller (PD controller)

Output signal $u(t)$ is proportional to error signal $e(t)$ & the derivative of error signal

$$u(t) \propto \left[e(t) + \frac{d}{dt} e(t) \right]$$

$$\therefore u(t) = K_p e(t) + K_d \frac{d}{dt} e(t)$$

K_p - proportional gain

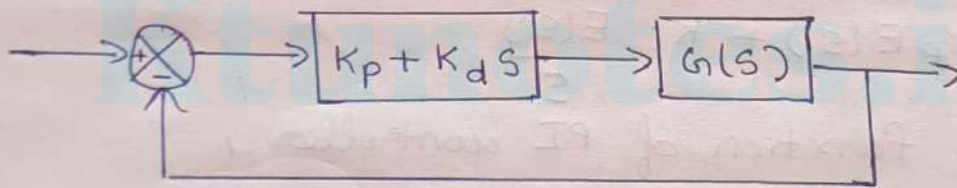
K_d - Derivative constant or gain

Taking Laplace transform,

$$U(s) = K_p E(s) + K_d S \cdot E(s)$$

$\therefore$ Transfer function of PD controller,

$$G_c(s) = \frac{U(s)}{E(s)} = K_p + K_d S$$



PD controller introduces a zero in the s/m & it $\uparrow^s$ the damping ratio. This results in the reduction of peak overshoot.

4. Proportional plus Integral plus Derivative controller. (PID controller)

$$u(t) \propto \left[e(t) + \int e(t) dt + \frac{d}{dt} e(t) \right]$$

$$u(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{d}{dt} e(t)$$

Taking Laplace transform,

$$U(s) = K_p E(s) + K_i \frac{E(s)}{s} + K_d s \cdot E(s)$$

∴ Transfer function of PID controller,

$$G_c(s) = \frac{U(s)}{E(s)} = K_p + \frac{K_i}{s} + K_d s$$

The PID controllers have the combined effect of all the three controllers — proportional, derivative & integral control actions. Hence the introduction of PID controller stabilises the gain, reduces the steady state error & peak overshoot of the s/m

